Computer Org. and Arch-Stallings-10Ed-Solution Manual-Ch1-10

logical design (الجامعة الإسلامية)

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SOLUTIONS MANUAL

COMPUTER ORGANIZATION AND ARCHITECTURE: DESIGNING FOR PERFORMANCE TENTH EDITION

CHAPTERS 1–10

WILLIAM STALLINGS



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NOTICE

This manual contains solutions to the review questions and homework problems in *Computer Organization and Architecture, Tenth Edition*. If you spot an error in a solution or in the wording of a problem, I would greatly appreciate it if you would forward the information via email to wllmst@me.net. An errata sheet for this manual, if needed, is available at http://www.box.net/shared/q4a7bmmtyc . File name is S-COA10e-mmyy

W.S.

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TABLE OF CONTENTS

Chapter 1 Basic Concepts and Computer Evolution 5
Chapter 2 Performance Issues 10
Chapter 3 Computer Function and Interconnection 19
Chapter 4 Cache Memory 25
Chapter 5 Internal Memory 40
Chapter 6 External Memory 49
Chapter 7 Input/Output 55
Chapter 8 Operating System Support 63
Chapter 9 Number Systems 71
Chapter 10 Computer Arithmetic 73

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CHAPTER 1 BASIC CONCEPTS AND COMPUTER EVOLUTION

ANSWERS TO QUESTIONS

1.1 **Computer architecture** refers to those attributes of a system visible to a programmer or, put another way, those attributes that have a direct impact on the logical execution of a program. **Computer organization** refers to the operational units and their interconnections that realize the architectural specifications. Examples of architectural attributes include the instruction set, the number of bits used to represent various data types (e.g., numbers, characters), I/O mechanisms, and techniques for addressing memory. Organizational attributes include those hardware details transparent to the programmer, such as control signals; interfaces between the computer and peripherals; and the memory technology used.

1.2 Computer structure refers to the way in which the components of a computer are interrelated. Computer function refers to the operation of each individual component as part of the structure.

1.3 Data processing; data storage; data movement; and control.

1.4 **Central processing unit (CPU):** Controls the operation of the computer and performs its data processing functions; often simply referred to as processor.
Main memory: Stores data.
I/O: Moves data between the computer and its external environment.
System interconnection: Some mechanism that provides for communication among CPU, main memory, and I/O. A common example of system interconnection is by means of a system bus, consisting of a number of conducting wires to which all the other components attach.

1.5 **Control unit:** Controls the operation of the CPU and hence the computer
Arithmetic and logic unit (ALU): Performs the computer's data processing functions
Registers: Provides storage internal to the CPU

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CPU interconnection: Some mechanism that provides for communication among the control unit, ALU, and registers

1.6 In a stored program computer, programs are represented in a form suitable for storing in memory alongside the data. The computer gets its instructions by reading them from memory, and a program can be set or altered by setting the values of a portion of memory.

1.7 Moore observed that the number of transistors that could be put on a single chip was doubling every year and correctly predicted that this pace would continue into the near future.

1.8 **Similar or identical instruction set:** In many cases, the same set of machine instructions is supported on all members of the family. Thus, a program that executes on one machine will also execute on any other. **Similar or identical operating system:** The same basic operating system is available for all family members. **Increasing speed:** The rate of instruction execution increases in going from lower to higher family members. **Increasing Number of I/O ports:** In going from lower to higher family members. **Increasing memory size:** In going from lower to higher family members. **Increasing cost:** In going from lower to higher family members.

1.9 In a microprocessor, all of the components of the CPU are on a single chip.

ANSWERS TO PROBLEMS

2.1 a

Location	Instruction/Value	Comments
0	<>	Constant (N) [initialized to some value]
1	1	Constant; Integer value = 1
2	2	Constant; Integer value = 2
3	0	Variable Y (initialized to integer zero); Sum(Y)
4L	LOAD M(0	N → AC
4R	ADD M(1)	AC + 1 → AC
5L	MUL M(0)	N(N+1) → AC
5R	DIV M(2)	AC/2 → AC
6L	STOR M(3)	AC → Y; saving the Sum in variable Y
6R	JUMP M(6,20:39)	Done; HALT

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b.

Location	Instruction/Value	Comments
0	<>	Constant (N) [initialized to some value]
1	1	Constant (loop counter increment)
2	1	Variable i (loop index value; current)
3	1	Variable Y = Sum of X values (Initialized to One)
4L	LOAD M(0	N → AC (the max limit)
4R	SUB M(2)	Compute N–i → AC
5L	JUMP + M(6,0:19)	Check AC > 0 ? [i < N]
5R	JUMP + M(5,20:39)	i=N; done so HALT
6L	LOAD M(2)	i<N so continue; Get loop counter i
6R	ADD M(1)	i+1 in AC
7L	STOR M(2)	AC → i
8R	ADD M(3)	i + Y in AC
8L	STOR M(3)	AC → Y
8R	JUMP M(4,0:19)	Continue at instruction located at address 4L

2.2 a.

Opcode	Operand
00000001	000000000010

b. First, the CPU must make access memory to fetch the instruction. The instruction contains the address of the data we want to load. During the execute phase accesses memory to load the data value located at that address for a total of two trips to memory.

2.3 To read a value from memory, the CPU puts the address of the value it wants into the MAR. The CPU then asserts the Read control line to memory and places the address on the address bus. Memory places the contents of the memory location passed on the data bus. This data is then transferred to the MBR. To write a value to memory, the CPU puts the address of the value it wants to write into the MAR. The CPU also places the data it wants to write into the MBR. The CPU then asserts the Write control line to memory and places the address on the address bus and the data on the data bus. Memory transfers the data on the data bus into the corresponding memory location.

2.4

Address	Contents
08A	LOAD M(0FA)
	STOR M(0FB)
08B	LOAD M(0FA)

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```
        JUMP +M(08D)
08C     LOAD –M(0FA)
        STOR M(0FB)
08D
```

This program will store the absolute value of content at memory location 0FA into memory location 0FB.

2.5 All data paths to/from MBR are 40 bits. All data paths to/from MAR are 12 bits. Paths to/from AC are 40 bits. Paths to/from MQ are 40 bits.

2.6 The purpose is to increase performance. When an address is presented to a memory module, there is some time delay before the read or write operation can be performed. While this is happening, an address can be presented to the other module. For a series of requests for successive words, the maximum rate is doubled.

2.7 The discrepancy can be explained by noting that other system components aside from clock speed make a big difference in overall system speed. In particular, memory systems and advances in I/O processing contribute to the performance ratio. A system is only as fast as its slowest link. In recent years, the bottlenecks have been the performance of memory modules and bus speed.

2.8 As noted in the answer to Problem 2.7, even though the Intel machine may have a faster clock speed (2.4 GHz vs. 1.2 GHz), that does not necessarily mean the system will perform faster. Different systems are not comparable on clock speed. Other factors such as the system components (memory, buses, architecture) and the instruction sets must also be taken into account. A more accurate measure is to run both systems on a benchmark. Benchmark programs exist for certain tasks, such as running office applications, performing floating-point operations, graphics operations, and so on. The systems can be compared to each other on how long they take to complete these tasks. According to Apple Computer, the G4 is comparable or better than a higher-clock speed Pentium on many benchmarks.

2.9 This representation is wasteful because to represent a single decimal digit from 0 through 9 we need to have ten tubes. If we could have an arbitrary number of these tubes ON at the same time, then those same tubes could be treated as binary bits. With ten bits, we can represent 2^{10} patterns, or 1024 patterns. For integers, these patterns could be used to represent the numbers from 0 through 1023.

1.10 a. No. These programs are never considered to be embedded because they are not an integral component of a larger system.

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b. Yes, regardless of what the disk drive is used for. The software (firmware, actually) within the disk drive controls the HDA (head disk assembly) hardware and is hard realtime as well.

c. No, because that computer may be a general-purpose computer that is not part of a larger system.

d. No. People often say that PDAs are embedded because they are very small and constrained and because PDA OS and application software is kept in non-volatile memory, but PDAs parallel the desktop systems used to run office productivity applications, and no special hardware is being controlled.

e. Yes. The firmware in the cell phone is controlling the radio hardware.

f. Yes. These computers were generally some of the most powerful computers available when the system was built, are located in a large computer room occupying almost one whole floor of a building, and may be hundreds of meters away from the radar hardware. However, the software running in these computers controls the radar hardware; therefore, the computers are an integral component of a larger system.

g. If the FMS is not connected to the avionics and is used only for logistics computations, a function readily performed on a laptop, then the FMS is clearly not embedded.

h. Yes, both in the simulator, and in the thing being tested in the HIL simulator. Hardware is being controlled on both sides.

i. Yes. In this case the "system" is the combination of the pacemaker and the person's heart.

j. Yes. It is part of a larger system, the engine, and it is directly monitoring and controlling the engine through special hardware.

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CHAPTER 2 PERFORMANCE ISSUES

ANSWERS TO QUESTIONS

2.1 ■ Pipelining: The execution of an instruction involves multiple stages of operation, including fetching the instruction, decoding the opcode, fetching operands, performing a calculation, and so on.
■ Branch prediction: The processor looks ahead in the instruction code fetched from memory and predicts which branches, or groups of instructions, are likely to be processed next.
■ Superscalar execution: This is the ability to issue more than one instruction in every processor clock cycle.
■ Data flow analysis: The processor analyzes which instructions are dependent on each other's results, or data, to create an optimized schedule of instructions.
■ Speculative execution: Using branch prediction and data flow analysis, some processors speculatively execute instructions ahead of their actual appearance in the program execution, holding the results in temporary locations.

2.2 Performance balance is the adjustment/tuning of the organization and architecture to compensate for the mismatch among the capabilities of the various components.

2.3 Multicore: the use of multiple processing units, called cores, on a single chip.
Many integrated core (MIC): a chip containing a large number (50 or more) cores.
General-purpose computing on GPUs (GPGPU): A GPU designed to support a broad range of applications.

2.4 Amdahl's law deals with the potential speedup of a program using multiple processors compared to a single processor. The law indicates the amount of speedup as a function of the fraction of code that can be executed in parallel.

2.5 Little's law applies to queuing systems. It defines the relationship between arrival rate, average time spent in system, and the average number of items in the system.

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2.6 MIPS = millions of instruction executions per second. FLOPS = floating-point operations per second.

2.7 Arithmetic mean: sum of a set of values divided by the number of values.
Geometric mean: the nth root of the product of a set of n values.
Harmonic mean: for n values, n divided by the product of the reciprocal of the n values.

2.8 1. It is written in a high-level language, making it portable across different machines.
2. It is representative of a particular kind of programming domain or paradigm, such as systems programming, numerical programming, or commercial programming.
3. It can be measured easily.
4. It has wide distribution.

2.9 This is a collection of benchmark suites is defined and maintained by the Standard Performance Evaluation Corporation (SPEC).

2.10 ■Base metric: These are required for all reported results and have strict guidelines for compilation. In essence, the standard compiler with more or less default settings should be used on each system under test to achieve comparable results.
■Peak metric: This enables users to attempt to optimize system performance by optimizing the compiler output.
■Speed metric: This is simply a measurement of the time it takes to execute a compiled benchmark.
■Rate metric: This is a measurement of how many tasks a computer can accomplish in a certain amount of time; this is called a throughput, capacity or rate measure.

ANSWERS TO PROBLEMS

2.1 *CPI* = 1.55; MIPS rate = 25.8; Execution time = 3.87 ms.

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2.2 a.

$$CPI_A = \frac{\sum CPI_i \times I_i}{I_c} = \frac{(8\times 1 + 4\times 3 + 2\times 4 + 4\times 3)\times 10^6}{(8+4+2+4)\times 10^6} \approx 2.22$$

$$MIPS_A = \frac{f}{CPI_A \times 10^6} = \frac{200\times 10^6}{2.22\times 10^6} = 90$$

$$CPU_A = \frac{I_c \times CPI_A}{f} = \frac{18\times 10^6 \times 2.2}{200\times 10^6} = 0.2 \text{ s}$$

$$CPI_B = \frac{\sum CPI_i \times I_i}{I_c} = \frac{(10\times 1 + 8\times 2 + 2\times 4 + 4\times 3)\times 10^6}{(10+8+2+4)\times 10^6} \approx 1.92$$

$$MIPS_B = \frac{f}{CPI_B \times 10^6} = \frac{200\times 10^6}{1.92\times 10^6} = 104$$

$$CPU_B = \frac{I_c \times CPI_B}{f} = \frac{24\times 10^6 \times 1.92}{200\times 10^6} = 0.23 \text{ s}$$

b. Although machine B has a higher MIPS than machine A, it requires a longer CPU time to execute the same set of benchmark programs.

2.3 **a.** We can express the MIPs rate as: [(MIPS rate)/10^6] = I_c/T. So that: $I_c = T \times$ [(MIPS rate)/10^6]. The ratio of the instruction count of the RS/6000 to the VAX is [x × 18]/[12x × 1] = 1.5.

b. For the Vax, *CPI* = (5 MHz)/(1 MIPS) = 5.
For the RS/6000, *CPI* = 25/18 = 1.39.

2.4 From Equation (2.3), MIPS = $I_c/(T \times 10^6)$ = 100/*T*. The MIPS values are:

	Computer A	**Computer B**	**Computer C**
Program 1	100	10	5
Program 2	0.1	1	5
Program 3	0.2	0.1	2
Program 4	1	0.125	1

	Arithmetic mean	**Rank**	**Harmonic mean**	**Rank**
Computer A	25.325	1	0.25	2
Computer B	2.8	3	0.21	3
Computer C	3.25	2	2.1	1

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2.5 **a.** Normalized to R:

Benchmark	Processor		
	R	M	Z
E	1.00	1.71	3.11
F	1.00	1.19	1.19
H	1.00	0.43	0.49
I	1.00	1.11	0.60
K	1.00	2.10	2.09
Arithmetic mean	1.00	1.31	1.50

b. Normalized to M:

Benchmark	Processor		
	R	M	Z
E	0.59	1.00	1.82
F	0.84	1.00	1.00
H	2.32	1.00	1.13
I	0.90	1.00	0.54
K	0.48	1.00	1.00
Arithmetic mean	1.01	1.00	1.10

c. Recall that the larger the ratio, the higher the speed. Based on (a) R is the slowest machine, by a significant amount. Based on (b), M is the slowest machine, by a modest amount.

d. Normalized to R:

Benchmark	Processor		
	R	M	Z
E	1.00	1.71	3.11
F	1.00	1.19	1.19
H	1.00	0.43	0.49
I	1.00	1.11	0.60
K	1.00	2.10	2.09
Geometric mean	1.00	1.15	1.18

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Normalized to M:

Benchmark	Processor		
	R	M	Z
E	0.59	1.00	1.82
F	0.84	1.00	1.00
H	2.32	1.00	1.13
I	0.90	1.00	0.54
K	0.48	1.00	1.00
Geometric mean	0.87	1.00	1.02

Using the geometric mean, R is the slowest no matter which machine is used for normalization.

2.6 a. Normalized to X:

Benchmark	Processor		
	X	Y	Z
1	1	2.0	0.5
2	1	0.5	2.0
Arithmetic mean	1	1.25	1.25
Geometric mean	1	1	1

Normalized to Y:

Benchmark	Processor		
	X	Y	Z
1	0.5	1	0.25
2	2.0	1	4.0
Arithmetic mean	1.25	1	2.125
Geometric mean	1	1	1

Machine Y is twice as fast as machine X for benchmark 1, but half as fast for benchmark 2. Similarly machine Z is half as fast as X for benchmark 1, but twice as fast for benchmark 2. Intuitively, these three machines have equivalent performance. However, if we

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normalize to X and compute the arithmetic mean of the speed metric, we find that Y and Z are 25% faster than X. Now, if we normalize to Y and compute the arithmetic mean of the speed metric, we find that X is 25% faster than Y and Z is more than twice as fast as Y. Clearly, the arithmetic mean is worthless in this context.

b. When the geometric mean is used, the three machines are shown to have equal performance when normalized to X, and also equal performance when normalized to Y. These results are much more in line with our intuition.

2.7 a. Assuming the same instruction mix means that the additional instructions for each task should be allocated proportionally among the instruction types. So we have the following table:

Instruction Type	CPI	Instruction Mix
Arithmetic and logic	1	60%
Load/store with cache hit	2	18%
Branch	4	12%
Memory reference with cache miss	12	10%

$CPI = 0.6 + (2 \times 0.18) + (4 \times 0.12) + (12 \times 0.1) = 2.64$. The CPI has increased due to the increased time for memory access.

b. $MIPS = 400/2.64 = 152$. There is a corresponding drop in the MIPS rate.

c. The speedup factor is the ratio of the execution times. Using Equation 2.2, we calculate the execution time as $T = I_c/(MIPS \times 10^6)$.

For the single-processor case, $T_1 = (2 \times 10^6)/(178 \times 10^6) = 11$ ms.

With 8 processors, each processor executes 1/8 of the 2 million instructions plus the 25,000 overhead instructions. For this case, the execution time for each of the 8 processors is

$$T_8 = \frac{\frac{2\times10^6}{8} + 0.025\times10^6}{152\times10^6} = 1.8 \text{ ms}$$

Therefore we have

$$\text{Speedup} = \frac{\text{time to execute program on a single processor}}{\text{time to execute program on } N \text{ parallel processors}} = \frac{11}{1.8} = 6.11$$

d. The answer to this question depends on how we interpret Amdahl's' law. There are two inefficiencies in the parallel system. First, there are additional instructions added to coordinate between threads. Second, there is contention for memory access. The way that the problem is stated implies that none of the code is inherently serial. All of it is parallelizable, but with scheduling overhead. One could argue that the memory access conflict means that to some extent

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memory reference instructions are not parallelizable. But based on the information given, it is not clear how to quantify this effect in Amdahl's equation. If we assume that the fraction of code that is parallelizable is $f = 1$, then Amdahl's law reduces to *Speedup* = $N = 8$ for this case. Thus the actual speedup is only about 75% of the theoretical speedup.

2.8 a. Speedup = (time to access in main memory)/(time to access in cache) = T_2/T_1.

b. The average access time can be computed as $T = H \times T_1 + (1 - H) \times T_2$

Using Equation (2.8):

$$\text{Speedup} = \frac{\text{Execution time before enhancement}}{\text{Execution time after enhancement}} = \frac{T_2}{T} = \frac{T_2}{H \times T_1 + (1-H)T_2} = \frac{1}{(1-H) + H\frac{T_1}{T_2}}$$

c. $T = H \times T_1 + (1 - H) \times (T_1 + T_2) = T_1 + (1 - H) \times T_2)$

This is Equation (4.2) in Chapter 4. Now,

$$\text{Speedup} = \frac{\text{Execution time before enhancement}}{\text{Execution time after enhancement}} = \frac{T_2}{T} = \frac{T_2}{T_1 + (1-H)T_2} = \frac{1}{(1-H) + \frac{T_1}{T_2}}$$

In this case, the denominator is larger, so that the speedup is less.

2.9 $T_w = w/\lambda = 8/18 = 0.44$ hours

2.10 a. A = CW

b. A = LT

c. CW = LT $\Rightarrow$ L = W(C/T)
but λ = C/T

2.11 a. Number the rectangles from bottom to top:

Rectangle	Width	Height	Area
1	2	1	2
2	2	1	2
3	2	1	2
4	1	1	1
5	3	1	3
6	1	1	1
		Total	11

b. Number the rectangles from left to right:

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Rectangle	Width	Height	Area
1	1	1	1
2	.5	2	1
3	.5	3	1.5
4	1	2	2
5	0.25	1	0.25
6	0.25	2	0.5
7	0.75	1	0.75
8	2.5	1	2.5
9	0.5	2	1
10	0.5	1	.5
		Total	11

2.12 a. We have FLOPSref = I/Tref, FLOPSsut = I/Tsut
So Tref = I/FLOPSref, Tsut = I/FLOPSsut. Therefore:

$$r_i = \frac{Tref_i}{Tsut_i} = \frac{I_i / FLOPSref_i}{I_i / FLOPSsut_i} = \frac{FLOPSsut_i}{FLOPSref_i}$$

b. $\frac{FLOPSref_i - FLOPSsut_i}{FLOPSsut_i}$ $\frac{Tref_i - Tsut_i}{Tsut_i}$

2.13

Machine	Execution time	Speedup	Relative Change
A	480	1	0
B	360	1.33	+0.33
C	540	0.89	–0.11
D	210	2.29	+1.29

Looking at column 2, it is easy to see that system D is the fastest system, followed by B, A, and C. Column 3 gives a more precise indication of exactly how much fast one system is than the others. Column 4 provides the same level of precision and clarity as column 3.

2.14

Machine	Execution time	Speedup	Relative Change
D	210	1	0
A	480	0.44	–0.56
B	360	0.58	–0.42
C	540	0.39	–0.61

No change to the relative rankings.

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2.15

	Computer A time (secs)	Computer B time (secs)	Computer C time (secs)	Computer A rate (MFLOPS)	Computer B rate (MFLOPS)	Computer C rate (MFLOPS)
Program 1 (10^8 FP ops)	2.0	1.0	0.2	50	100	500
Program 2 (10^8 FP ops)	4	2.0	4.0	250	50	25
Total execution time	2.4	3.0	4.2			
Arithmetic mean of times	1.2	1.5	2.1			
Inverse of total execution time (1/sec)	0.42	0.33	0.24			
Arithmetic mean of rates				150	75.00	262.5
Harmonic mean of rates				83.33	66.67	47.6

Again, the arithmetic mean of rates does not reflect the speed ranks, whereas the harmonic mean does.

2.16 a. We use two identities: $e^X e^Y = e^{(X + Y)}$ and $\ln(X^p) = p \ln x$

$$e^{\left(\frac{1}{n}\sum_{i=1}^{n}\ln(x_i)\right)} = e^{\left(\sum_{i=1}^{n}\frac{\ln(x_i)}{n}\right)} = \prod_{i=1}^{n} e^{\frac{\ln(x_i)}{n}} = \prod_{i=1}^{n} e^{\ln\left((x_i)^{1/n}\right)} = \prod_{i=1}^{n} e^{\ln\left((x_i)^{1/n}\right)} = \prod_{i=1}^{n}\left((x_i)^{1/n}\right) = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

b. The intermediate values in the calculation are not as large.

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CHAPTER 3 COMPUTER FUNCTION AND INTERCONNECTION

ANSWERS TO QUESTIONS

3.1 **Processor-memory:** Data may be transferred from processor to memory or from memory to processor. **Processor-I/O:** Data may be transferred to or from a peripheral device by transferring between the processor and an I/O module. **Data processing:** The processor may perform some arithmetic or logic operation on data. **Control:** An instruction may specify that the sequence of execution be altered.

3.2 **Instruction address calculation (iac):** Determine the address of the next instruction to be executed. **Instruction fetch (if):** Read instruction from its memory location into the processor. **Instruction operation decoding (iod):** Analyze instruction to determine type of operation to be performed and operand(s) to be used. **Operand address calculation (oac):** If the operation involves reference to an operand in memory or available via I/O, then determine the address of the operand. **Operand fetch (of):** Fetch the operand from memory or read it in from I/O. **Data operation (do):** Perform the operation indicated in the instruction. **Operand store (os):** Write the result into memory or out to I/O.

3.3 **(1)** Disable all interrupts while an interrupt is being processed. **(2)** Define priorities for interrupts and to allow an interrupt of higher priority to cause a lower-priority interrupt handler to be interrupted.

3.4 **Memory to processor:** The processor reads an instruction or a unit of data from memory. **Processor to memory:** The processor writes a unit of data to memory. **I/O to processor:** The processor reads data from an I/O device via an I/O module. **Processor to I/O:** The processor sends data to the I/O device. **I/O to or from memory:** For these two cases, an I/O module is allowed to exchange data directly with memory, without going through the processor, using direct memory access (DMA).

3.5 **Physical layer:** Consists of the actual wires carrying the signals, as well as circuitry and logic to support ancillary features required in the

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transmission and receipt of the 1s and 0s. The unit of transfer at the Physical layer is 20 bits, which is called a **Phit** (physical unit).
Link layer: Responsible for reliable transmission and flow control. The Link layer's unit of transfer is an 80-bit **Flit** (flow control unit).
Routing layer: Provides the framework for directing packets through the fabric.
Protocol layer: The high-level set of rules for exchanging **packets** of data between devices. A packet is comprised of an integral number of Flits.

3.6 **Physical layer:** Consists of the actual wires carrying the signals, as well as circuitry and logic to support ancillary features required in the transmission and receipt of the 1s and 0s.
Data link layer: Is responsible for reliable transmission and flow control. Data packets generated and consumed by the DLL are called Data Link Layer Packets (DLLPs).
Transaction layer: Generates and consumes data packets used to implement load/store data transfer mechanisms and also manages the flow control of those packets between the two components on a link. Data packets generated and consumed by the TL are called Transaction Layer Packets (TLPs).

ANSWERS TO PROBLEMS

3.1 Memory (contents in hex): 300: 3005; 301: 5940; 302: 7006
Step 1: 3005 → IR; **Step 2:** 3 → AC
Step 3: 5940 → IR; **Step 4:** 3 + 2 = 5 → AC
Step 5: 7006 → IR; **Step 6:** AC → Device 6

3.2
1. a. The PC contains 300, the address of the first instruction. This value is loaded in to the MAR.
b. The value in location 300 (which is the instruction with the value 1940 in hexadecimal) is loaded into the MBR, and the PC is incremented. These two steps can be done in parallel.
c. The value in the MBR is loaded into the IR.
2. a. The address portion of the IR (940) is loaded into the MAR.
b. The value in location 940 is loaded into the MBR.
c. The value in the MBR is loaded into the AC.
3. a. The value in the PC (301) is loaded in to the MAR.
b. The value in location 301 (which is the instruction with the value 5941) is loaded into the MBR, and the PC is incremented.
c. The value in the MBR is loaded into the IR.
4. a. The address portion of the IR (941) is loaded into the MAR.
b. The value in location 941 is loaded into the MBR.

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c. The old value of the AC and the value of location MBR are added and the result is stored in the AC.

5. a. The value in the PC (302) is loaded in to the MAR.
b. The value in location 302 (which is the instruction with the value 2941) is loaded into the MBR, and the PC is incremented.
c. The value in the MBR is loaded into the IR.

6. a. The address portion of the IR (941) is loaded into the MAR.
b. The value in the AC is loaded into the MBR.
c. The value in the MBR is stored in location 941.

3.3 a. 2^{24} = 16 MBytes

b. (1) If the local address bus is 32 bits, the whole address can be transferred at once and decoded in memory. However, because the data bus is only 16 bits, it will require 2 cycles to fetch a 32-bit instruction or operand.

(2) The 16 bits of the address placed on the address bus can't access the whole memory. Thus a more complex memory interface control is needed to latch the first part of the address and then the second part (because the microprocessor will end in two steps). For a 32-bit address, one may assume the first half will decode to access a "row" in memory, while the second half is sent later to access a "column" in memory. In addition to the two-step address operation, the microprocessor will need 2 cycles to fetch the 32-bit instruction/operand.

c. The program counter must be at least 24 bits. Typically, a 32-bit microprocessor will have a 32-bit external address bus and a 32-bit program counter, unless on-chip segment registers are used that may work with a smaller program counter. If the instruction register is to contain the whole instruction, it will have to be 32-bits long; if it will contain only the op code (called the op code register) then it will have to be 8 bits long.

3.4 In cases **(a)** and **(b)**, the microprocessor will be able to access 2^{16} = 64K bytes; the only difference is that with an 8-bit memory each access will transfer a byte, while with a 16-bit memory an access may transfer a byte or a 16-byte word. For case **(c)**, separate input and output instructions are needed, whose execution will generate separate "I/O signals" (different from the "memory signals" generated with the execution of memory-type instructions); at a minimum, one additional output pin will be required to carry this new signal. For case **(d)**, it can support 2^8 = 256 input and 2^8 = 256 output byte ports and the same number of input and output 16-bit ports; in either case, the distinction between an input and an output port is defined by the different signal that the executed input or output instruction generated.

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3.5 Clock cycle = $\frac{1}{8\,\text{MHz}} = 125\,\text{ns}$

Bus cycle = 4×125 ns = 500 ns
2 bytes transferred every 500 ns; thus transfer rate = 4 MBytes/sec

Doubling the frequency may mean adopting a new chip manufacturing technology (assuming each instructions will have the same number of clock cycles); doubling the external data bus means wider (maybe newer) on-chip data bus drivers/latches and modifications to the bus control logic. In the first case, the speed of the memory chips will also need to double (roughly) not to slow down the microprocessor; in the second case, the "wordlength" of the memory will have to double to be able to send/receive 32-bit quantities.

3.6 a. Input from the Teletype is stored in INPR. The INPR will only accept data from the Teletype when FGI=0. When data arrives, it is stored in INPR, and FGI is set to 1. The CPU periodically checks FGI. If FGI =1, the CPU transfers the contents of INPR to the AC and sets FGI to 0.

When the CPU has data to send to the Teletype, it checks FGO. If FGO = 0, the CPU must wait. If FGO = 1, the CPU transfers the contents of the AC to OUTR and sets FGO to 0. The Teletype sets FGI to 1 after the word is printed.

b. The process described in **(a)** is very wasteful. The CPU, which is much faster than the Teletype, must repeatedly check FGI and FGO. If interrupts are used, the Teletype can issue an interrupt to the CPU whenever it is ready to accept or send data. The IEN register can be set by the CPU (under programmer control)

3.7 a. During a single bus cycle, the 8-bit microprocessor transfers one byte while the 16-bit microprocessor transfers two bytes. The 16-bit microprocessor has twice the data transfer rate.

b. Suppose we do 100 transfers of operands and instructions, of which 50 are one byte long and 50 are two bytes long. The 8-bit microprocessor takes 50 + (2 x 50) = 150 bus cycles for the transfer. The 16-bit microprocessor requires 50 + 50 = 100 bus cycles. Thus, the data transfer rates differ by a factor of 1.5.

3.8 The whole point of the clock is to define event times on the bus; therefore, we wish for a bus arbitration operation to be made each clock cycle. This requires that the priority signal propagate the length of the daisy chain (Figure 3.26) in one clock period. Thus, the maximum number of masters is determined by dividing the amount of time it takes a bus master to pass through the bus priority by the clock period.

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3.9 The lowest-priority device is assigned priority 16. This device must defer to all the others. However, it may transmit in any slot not reserved by the other SBI devices.

3.10 At the beginning of any slot, if none of the TR lines is asserted, only the priority 16 device may transmit. This gives it the lowest average wait time under most circumstances. Only when there is heavy demand on the bus, which means that most of the time there is at least one pending request, will the priority 16 device not have the lowest average wait time.

3.11 **a.** With a clocking frequency of 10 MHz, the clock period is 10^{-9} s = 100 ns. The length of the memory read cycle is 300 ns.

b. The Read signal begins to fall at 75 ns from the beginning of the third clock cycle (middle of the second half of T_3). Thus, memory must place the data on the bus no later than 55 ns from the beginning of T_3.

3.12 **a**. The clock period is 125 ns. Therefore, two clock cycles need to be inserted.

b. From Figure 3.19, the Read signal begins to rise early in T_2. To insert two clock cycles, the Ready line can be put in low at the beginning of T_2 and kept low for 250 ns.

3.13 **a.** A 5 MHz clock corresponds to a clock period of 200 ns. Therefore, the Write signal has a duration of 150 ns.

b. The data remain valid for 150 + 20 = 170 ns.

c. One wait state.

3.14 **a.** Without the wait states, the instruction takes 16 bus clock cycles. The instruction requires four memory accesses, resulting in 8 wait states. The instruction, with wait states, takes 24 clock cycles, for an increase of 50%.

b. In this case, the instruction takes 26 bus cycles without wait states and 34 bus cycles with wait states, for an increase of 33%.

3.15 **a.** The clock period is 125 ns. One bus read cycle takes 500 ns = 0.5 μs. If the bus cycles repeat one after another, we can achieve a data transfer rate of 2 MB/s.

b. The wait state extends the bus read cycle by 125 ns, for a total duration of 0.625 μs. The corresponding data transfer rate is 1/0.625 = 1.6 MB/s.

3.16 A bus cycle takes 0.25 μs, so a memory cycle takes 1 μs. If both operands are even-aligned, it takes 2 μs to fetch the two operands. If

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one is odd-aligned, the time required is 3 μs. If both are odd-aligned, the time required is 4 μs.

3.17 Consider a mix of 100 instructions and operands. On average, they consist of 20 32-bit items, 40 16-bit items, and 40 bytes. The number of bus cycles required for the 16-bit microprocessor is $(2 \times 20) + 40 + 40 = 120$. For the 32-bit microprocessor, the number required is 100. This amounts to an improvement of 20/120 or about 17%.

3.18 The processor needs another nine clock cycles to complete the instruction. Thus, the Interrupt Acknowledge will start after 900 ns.

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CHAPTER 4 CACHE MEMORY

ANSWERS TO QUESTIONS

4.1 **Sequential access:** Memory is organized into units of data, called records. Access must be made in a specific linear sequence. **Direct access:** Individual blocks or records have a unique address based on physical location. Access is accomplished by direct access to reach a general vicinity plus sequential searching, counting, or waiting to reach the final location. **Random access:** Each addressable location in memory has a unique, physically wired-in addressing mechanism. The time to access a given location is independent of the sequence of prior accesses and is constant.

4.2 Faster access time, greater cost per bit; greater capacity, smaller cost per bit; greater capacity, slower access time.

4.3 It is possible to organize data across a memory hierarchy such that the percentage of accesses to each successively lower level is substantially less than that of the level above. Because memory references tend to cluster, the data in the higher-level memory need not change very often to satisfy memory access requests.

4.4 In a cache system, **direct mapping** maps each block of main memory into only one possible cache line. **Associative mapping** permits each main memory block to be loaded into any line of the cache. In **set-associative mapping**, the cache is divided into a number of sets of cache lines; each main memory block can be mapped into any line in a particular set.

4.5 One field identifies a unique word or byte within a block of main memory. The remaining two fields specify one of the blocks of main memory. These two fields are a line field, which identifies one of the lines of the cache, and a tag field, which identifies one of the blocks that can fit into that line.

4.6 A tag field uniquely identifies a block of main memory. A word field identifies a unique word or byte within a block of main memory.

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4.7 One field identifies a unique word or byte within a block of main memory. The remaining two fields specify one of the blocks of main memory. These two fields are a set field, which identifies one of the sets of the cache, and a tag field, which identifies one of the blocks that can fit into that set.

4.8 **Spatial locality** refers to the tendency of execution to involve a number of memory locations that are clustered. **Temporal locality** refers to the tendency for a processor to access memory locations that have been used recently.

4.9 **Spatial locality** is generally exploited by using larger cache blocks and by incorporating prefetching mechanisms (fetching items of anticipated use) into the cache control logic. **Temporal locality** is exploited by keeping recently used instruction and data values in cache memory and by exploiting a cache hierarchy.

ANSWERS TO PROBLEMS

4.1 The cache is divided into 16 sets of 4 lines each. Therefore, 4 bits are needed to identify the set number. Main memory consists of 4K = 2^{12} blocks. Therefore, the set plus tag lengths must be 12 bits and therefore the tag length is 8 bits. Each block contains 128 words. Therefore, 7 bits are needed to specify the word.

	TAG	SET	WORD
Main memory address =	8	4	7

4.2 There are a total of 8 kbytes/16 bytes = 512 lines in the cache. Thus the cache consists of 256 sets of 2 lines each. Therefore 8 bits are needed to identify the set number. For the 64-Mbyte main memory, a 26-bit address is needed. Main memory consists of 64-Mbyte/16 bytes = 2^{22} blocks. Therefore, the set plus tag lengths must be 22 bits, so the tag length is 14 bits and the word field length is 4 bits.

	TAG	SET	WORD
Main memory address =	14	8	4

4.3

Address	111111	666666	BBBBBB
a. Tag/Line/Word	11/444/1	66/1999/2	BB/2EEE/3
b. Tag /Word	44444/1	199999/2	2EEEEE/3
c. Tag/Set/Word	22/444/1	CC/1999/2	177/EEE/3

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4.4 **a.** Address length: 24; number of addressable units: 2^{24}; block size: 4; number of blocks in main memory: 2^{22}; number of lines in cache: 2^{14}; size of tag: 8.

b. Address length: 24; number of addressable units: 2^{24}; block size: 4; number of blocks in main memory: 2^{22}; number of lines in cache: 4000 hex; size of tag: 22.

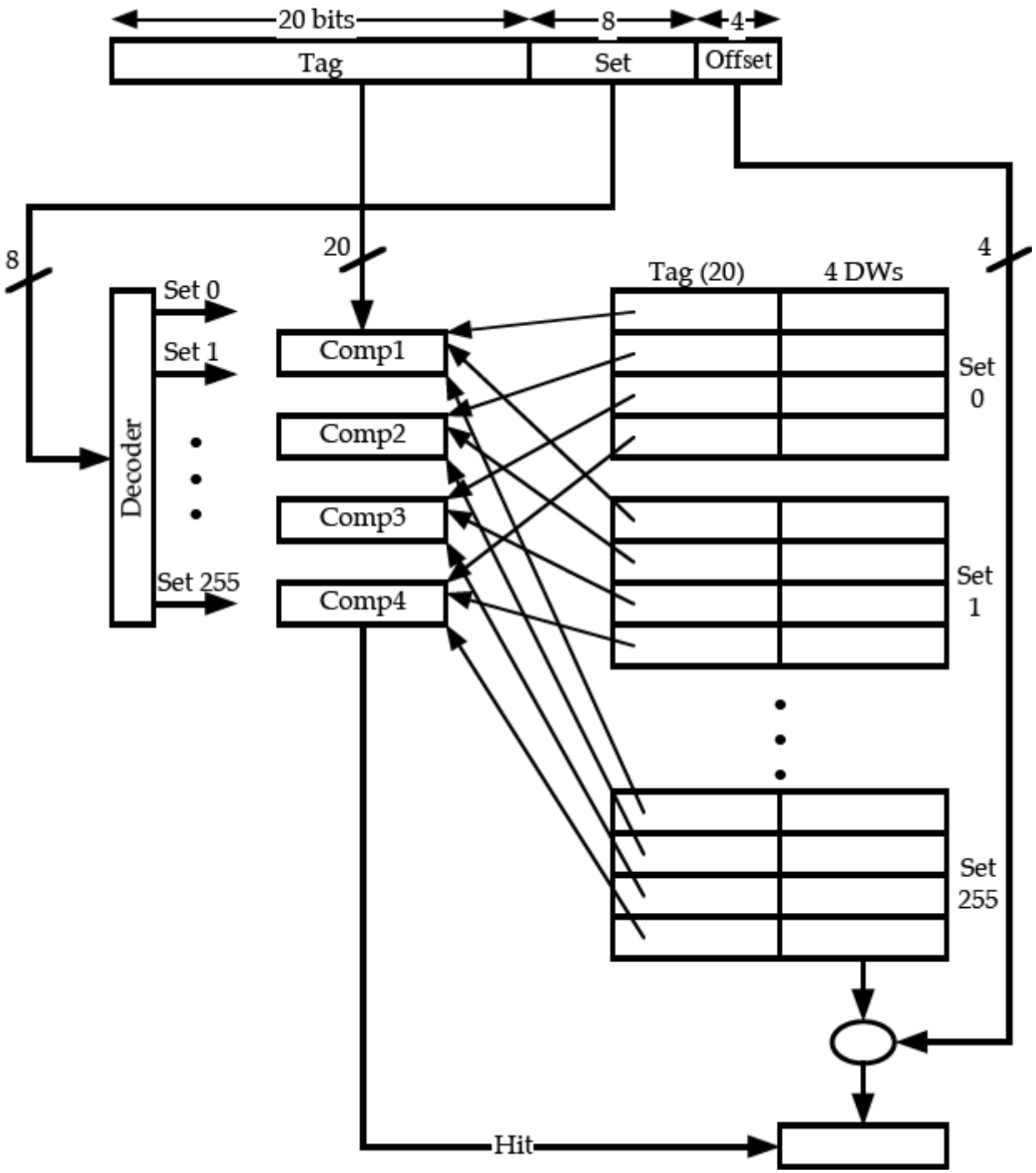
c. Address length: 24; number of addressable units: 2^{24}; block size: 4; number of blocks in main memory: 2^{22}; number of lines in set: 2; number of sets: 2^{13}; number of lines in cache: 2^{14}; size of tag: 9.

4.5 Block frame size = 16 bytes = 4 doublewords

Number of block frames in cache = $\frac{16\,\text{KBytes}}{16\,\text{Bytes}} = 1024$

Number of sets = $\frac{\text{Number of block frames}}{\text{Associativity}} = \frac{1024}{4} = 256\ \text{sets}$

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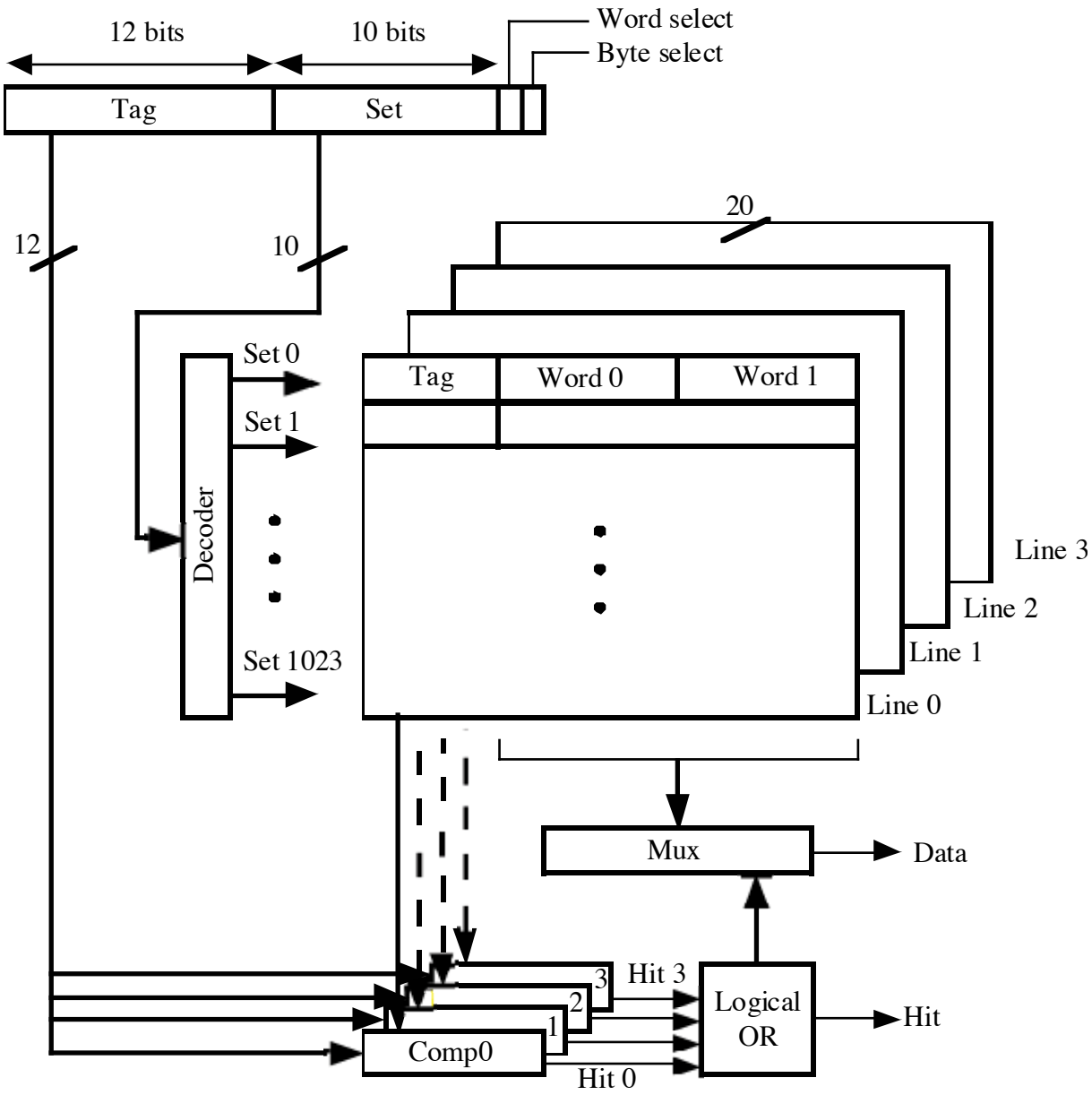
Example: doubleword from location ABCDE8F8 is mapped onto: set 143, any line, doubleword 2:

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4.6



4.7 A 32-bit address consists of a 21-bit tag field, a 7-bit set field, and a 4-bit word field. Each set in the cache includes 3 LRU bits and four lines. Each line consists of 4 32-bit words, a valid bit, and a 21-bit tag.

4.8 **a.** 8 leftmost bits = tag; 5 middle bits = line number; 3 rightmost bits = byte number

b. slot 3; slot 6; slot 3; slot 21

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c. Bytes with addresses 0001 1010 0001 1000 through 0001 1010 0001 1111 are stored in the cache

d. 256 bytes

e. Because two items with two different memory addresses can be stored in the same place in the cache. The tag is used to distinguish between them.

4.9 a. The bits are set according to the following rules with each access to the set:

1. If the access is to L0 or L1, B0 $\leftarrow$ 1.
2. If the access is to L0, B1 $\leftarrow$ 1.
3. If the access is to L1, B1 $\leftarrow$ 0.
4. If the access is to L2 or L3, B0 $\leftarrow$ 0.
5. If the access is to L2, B2 $\leftarrow$ 1.
6. If the access is to L3, B2 $\leftarrow$ 0.

The replacement algorithm works as follows (Figure 4.15): When a line must be replaced, the cache will first determine whether the most recent use was from L0 and L1 or L2 and L3. Then the cache will determine which of the pair of blocks was least recently used and mark it for replacement. When the cache is initialized or flushed all 128 sets of three LRU bits are set to zero.

b. The 80486 divides the four lines in a set into two pairs (L0, L1 and L2, L3). Bit B0 is used to select the pair that has been least-recently used. Within each pair, one bit is used to determine which member of the pair was least-recently used. However, the ultimate selection only approximates LRU. Consider the case in which the order of use was: L0, L2, L3, L1. The least-recently used pair is (L2, L3) and the least-recently used member of that pair is L2, which is selected for replacement. However, the least-recently used line of all is L0. Depending on the access history, the algorithm will always pick the least-recently used entry or the second least-recently used entry.

c. The most straightforward way to implement true LRU for a four-line set is to associate a two bit counter with each line. When an access occurs, the counter for that block is set to 0; all counters with values lower than the original value for the accessed block are incremented by 1. When a miss occurs and the set is not full, a new block is brought in, its counter is set to 0 and all other counters are incremented by 1. When a miss occurs and the set is full, the block with counter value 3 is replaced; its counter is set to 0 and all other counters are incremented by 1. This approach requires a total of 8 bits.

In general, for a set of N blocks, the above approach requires 2N bits. A more efficient scheme can be designed which requires only N(N–1)/2 bits. The scheme operates as follows. Consider a matrix R

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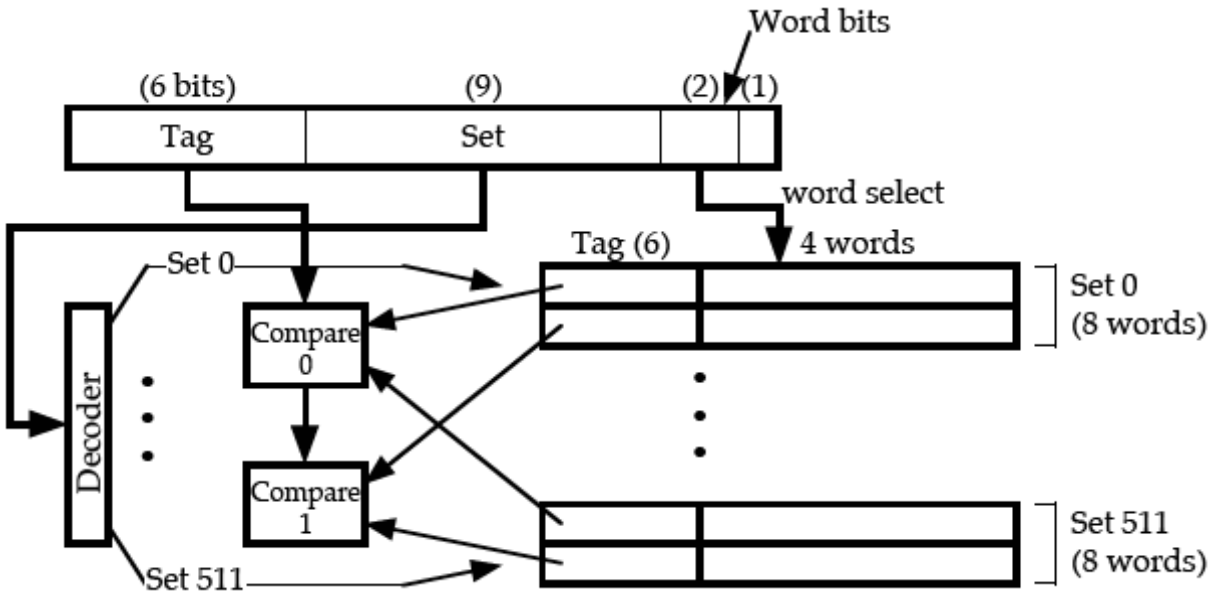
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with N rows and N columns, and take the upper-right triangular portion of the matrix, not counting the diagonal. For N = 4, we have the following layout:

	R(1,2)	R(1,3)	R(1,4)
		R(2,3)	R(2,4)
			R(3,4)

When line I is referenced, row I of R(I,J) is set to 1, and column I of R(J,I) is set to 0. The LRU block is the one for which the row is entirely equal to 0 (for those bits in the row; the row may be empty) and for which the column is entirely 1 (for all the bits in the column; the column may be empty). As can be seen for N = 4, a total of 6 bits are required.

4.10 Block size = 4 words = 2 doublewords; associativity K = 2; cache size = 4048 words; C = 1024 block frames; number of sets S = C/K = 512; main memory = 64K $\times$ 32 bits = 256 Kbytes = 2^{18} bytes; address = 18 bits.



4.11 **a.** Address format: Tag = 20 bits; Line = 6 bits; Word = 6 bits
Number of addressable units = $2^{s+w} = 2^{32}$ bytes; number of blocks in main memory = $2^s = 2^{26}$; number of lines in cache $2^r = 2^6 = 64$; size of tag = 20 bits.

b. Address format: Tag = 26 bits; Word = 6 bits

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Number of addressable units = $2^{s+w} = 2^{32}$ bytes; number of blocks in main memory = $2^s = 2^{26}$; number of lines in cache = undetermined; size of tag = 26 bits.

c. Address format: Tag = 9 bits; Set = 17 bits; Word = 6 bits
Number of addressable units = $2^{s+w} = 2^{32}$ bytes; Number of blocks in main memory = $2^s = 2^{26}$; Number of lines in set = $k = 4$; Number of sets in cache = $2^d = 2^{17}$; Number of lines in cache = $k \times 2^d = 2^{19}$; Size of tag = 9 bits.

4.12 a. Because the block size is 16 bytes and the word size is 1 byte, this means there are 16 words per block. We will need 4 bits to indicate which word we want out of a block. Each cache line/slot matches a memory block. That means each cache slot contains 16 bytes. If the cache is 64Kbytes then 64Kbytes/16 = 4096 cache slots. To address these 4096 cache slots, we need 12 bits (212 = 4096). Consequently, given a 20 bit (1 MByte) main memory address:

Bits 0-3 indicate the word offset (4 bits)
Bits 4-15 indicate the cache slot (12 bits)
Bits 16-19 indicate the tag (remaining bits)

F0010 = 1111 0000 0000 0001 0000
Word offset = 0000 = 0
Slot = 0000 0000 0001 = 001
Tag = 1111 = F

01234 = 0000 0001 0010 0011 0100
Word offset = 0100 = 4
Slot = 0001 0010 0011 = 123
Tag = 0000 = 0

CABBE = 1100 1010 1011 1011 1110
Word offset = 1110 = E
Slot = 1010 1011 1011 = ABB
Tag = 1100 = C

b. We need to pick any address where the slot is the same, but the tag (and optionally, the word offset) is different. Here are two examples where the slot is 1111 1111 1111

Address 1:
Word offset = 1111
Slot = 1111 1111 1111
Tag = 0000
Address = 0FFFF

Address 2:
Word offset = 0001
Slot = 1111 1111 1111
Tag = 0011
Address = 3FFF1

c. With a fully associative cache, the cache is split up into a TAG and a WORDOFFSET field. We no longer need to identify which slot a

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memory block might map to, because a block can be in any slot and we will search each cache slot in parallel. The word-offset must be 4 bits to address each individual word in the 16-word block. This leaves 16 bits leftover for the tag.

F0010

Word offset = 0h

Tag = F001h

CABBE

Word offset = Eh

Tag = CABBh

d. As computed in part a, we have 4096 cache slots. If we implement a two -way set associative cache, then it means that we put two cache slots into one set. Our cache now holds 4096/2 = 2048 sets, where each set has two slots. To address these 2048 sets we need 11 bits (211 = 2048). Once we address a set, we will simultaneously search both cache slots to see if one has a tag that matches the target. Our 20-bit address is now broken up as follows:

Bits 0-3 indicate the word offset

Bits 4-14 indicate the cache set

Bits 15-20 indicate the tag

F0010 = 1111 0000 0000 0001 0000

Word offset = 0000 = 0

Cache Set = 000 0000 0001 = 001

Tag = 11110 = 1 1110 = 1E

CABBE = 1100 1010 1011 1011 1110

Word offset = 1110 = E

Cache Set = 010 1011 1011 = 2BB

Tag = 11001 = 1 1001 = 19

4.13 Associate a 2-bit counter with each of the four blocks in a set. Initially, arbitrarily set the four values to 0, 1, 2, and 3 respectively. When a hit occurs, the counter of the block that is referenced is set to 0. The other counters in the set with values originally lower than the referenced counter are incremented by 1; the remaining counters are unchanged. When a miss occurs, the block in the set whose counter value is 3 is replaced and its counter set to 0. All other counters in the set are incremented by 1.

4.14 Writing back a line takes 30 + (7 $\times$ 5) = 65 ns, enough time for 2.17 single-word memory operations. If the average line that is written at least once is written more than 2.17 times, the write-back cache will be more efficient.

4.15 a. A reference to the first instruction is immediately followed by a reference to the second.

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b. The ten accesses to `a[i]` within the inner for loop which occur within a short interval of time.

4.16 Define

C_i = Average cost per bit, memory level i
S_i = Size of memory level i
T_i = Time to access a word in memory level i
H_i = Probability that a word is in memory i and in no higher-level memory
B_i = Time to transfer a block of data from memory level (i + 1) to memory level i

Let cache be memory level 1; main memory, memory level 2; and so on, for a total of N levels of memory. Then

$$C_s = \frac{\sum_{i=1}^{N} C_i S_i}{\sum_{i=1}^{N} S_i}$$

The derivation of T_S is more complicated. We begin with the result from probability theory that:

$$\text{Expected Value of } x = \sum_{i=1}^{N} i \Pr[x = 1]$$

We can write:

$$T_s = \sum_{i=1}^{N} T_i H_i$$

We need to realize that if a word is in M_1 (cache), it is read immediately. If it is in M_2 but not M_1, then a block of data is transferred from M_2 to M_1 and then read. Thus:

$$T_2 = B_1 + T_1$$

Further

$$T_3 = B_2 + T_2 = B_1 + B_2 + T_1$$

Generalizing:

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$$T_i = \sum_{j=1}^{i-1} B_j + T_1$$

So

$$T_s = \sum_{i=2}^{N} \sum_{j=1}^{i-1} \left(B_j H_i \right) + T_1 \sum_{i=1}^{N} H_i$$

But $$\sum_{i=1}^{N} H_i = 1$$

Finally

$$T_s = \sum_{i=2}^{N} \sum_{j=1}^{i-1} \left(B_j H_i \right) + T_1$$

4.17 Main memory consists of 512 blocks of 64 words. Cache consists of 16 sets; each set consists of 4 slots; each slot consists of 64 words. Locations 0 through 4351 in main memory occupy blocks 0 through 67. On the first fetch sequence, block 0 through 15 are read into sets 0 through 15; blocks 16 through 31 are read into sets 0 through 15; blocks 32-47 are read into sets 0 through 15; blocks 48-63 are read into sets 0 through 15; and blocks 64-67 are read into sets 0 through 3. Because each set has 4 slots, there is no replacement needed through block 63. The last 4 groups of blocks involve a replacement. On each successive pass, replacements will be required in sets 0 through 3, but all of the blocks in sets 4 through 15 remain undisturbed. Thus, on each successive pass, 48 blocks are undisturbed, and the remaining 20 must read in.

Let T be the time to read 64 words from cache. Then 10T is the time to read 64 words from main memory. If a word is not in the cache, then it can only be ready by first transferring the word from main memory to the cache and then reading the cache. Thus the time to read a 64-word block from cache if it is missing is 11T.

We can now express the improvement factor as follows. With no cache

Fetch time = (10 passes) (68 blocks/pass) (10T/block) = 6800T

With cache

Fetch time = (68) (11T) first pass
+ (9) (48) (T) + (9) (20) (11T) other passes
= 3160T

Improvement = $\frac{6800T}{3160T}$ = 2.15

4.18 **a.** Access 63 1 Miss Block 3 → Slot 3

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Access 64 1 Miss Block 4 → Slot 0
Access 65-70 6 Hits
Access 15 1 Miss Block 0 → Slot 0 First Loop
Access 16 1 Miss Block 1 → Slot 1
Access 17-31 15 Hits
Access 32 1 Miss Block 2 → Slot 2
Access 80 1 Miss Block 5 → Slot 1
Access 81-95 15 Hits
Access 15 1 Hit Second Loop
Access 16 1 Miss Block 1 → Slot 1
Access 17-31 15 hits
Access 32 1 Hit
Access 80 1 Miss Block 5 → Slot 1
Access 81-95 15 hits
Access 15 1 Hit Third Loop
Access 16 1 Miss Block 1 → Slot 1
Access 17-31 15 hits
Access 32 1 Hit
Access 80 1 Miss Block 5 → Slot 1
Access 81-95 15 hits
Access 15 1 Hit Fourth Loop
… Pattern continues to the Tenth Loop

For lines 63-70	2 Misses	6 Hits
First loop 15-32, 80-95	4 Misses	30 Hits
Second loop 15-32, 80-95	2 Misses	32 Hits
Third loop 15-32, 80-95	2 Misses	32 Hits
Fourth loop 15-32, 80-95	2 Misses	32 Hits
Fifth loop 15-32, 80-95	2 Misses	32 Hits
Sixth loop 15-32, 80-95	2 Misses	32 Hits
Seventh loop 15-32, 80-95	2 Misses	32 Hits
Eighth loop 15-32, 80-95	2 Misses	32 Hits
Ninth loop 15-32, 80-95	2 Misses	32 Hits
Tenth loop 15-32, 80-95	2 Misses	32 Hits
Total:	24 Misses	324 Hits

Hit Ratio = 324/348 = 0.931

b. Access 63 1 Miss Block 3 → Set 1 Slot 2
Access 64 1 Miss Block 4 → Set 0 Slot 0
Access 65-70 6 Hits
Access 15 1 Miss Block 0 → Set 0 Slot 1 First Loop
Access 16 1 Miss Block 1 → Set 1 Slot 3
Access 17-31 15 Hits
Access 32 1 Miss Block 2 → Set 0 Slot 0
Access 80 1 Miss Block 5 → Set 1 Slot 2
Access 81-95 15 Hits
Access 15 1 Hit Second Loop

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Access 16-31 16 Hits
Access 32 1 Hit
Access 80-95 16 Hits
… All hits for the next eight iterations

For lines 63-70	2 Misses	6 Hits
First loop 15-32, 80-95	4 Misses	30 Hits
Second loop 15-32, 80-95	0 Misses	34 Hits
Third loop 15-32, 80-95	0 Misses	34 Hits
Fourth loop 15-32, 80-95	0 Misses	34 Hits
Fifth loop 15-32, 80-95	0 Misses	34 Hits
Sixth loop 15-32, 80-95	0 Misses	34 Hits
Seventh loop 15-32, 80-95	0 Misses	34 Hits
Eighth loop 15-32, 80-95	0 Misses	34 Hits
Ninth loop 15-32, 80-95	0 Misses	34 Hits
Tenth loop 15-32, 80-95	0 Misses	34 Hits

Total = 6 Misses 342 Hits
Hit Ratio = 342/348 = 0.983

4.19 a. Cost = $C_m \times 8 \times 10^6 = 8 \times 10^3$ ¢ = \$80

b. Cost = $C_c \times 8 \times 10^6 = 8 \times 10^4$ ¢ = \$800

c. From Equation (4.1) : $1.1 \times T_1 = T_1 + (1 - H)T_2$

$(0.1)(100) = (1 - H)(1200)$

$H = 1190/1200$

4.20 a. Under the initial conditions, using Equation (4.1), the average access time is

$$T_1 + (1 - H)\, T_2 = 1 + (0.05)\, T_2$$

Under the changed conditions, the average access time is

$$1.5 + (0.03)\, T_2$$

For improved performance, we must have

$$1 + (0.05)\, T_2 > 1.5 + (0.03)\, T_2$$

Solving for T_2, the condition is $T_2 > 50$

b. As the time for access when there is a cache miss become larger, it becomes more important to increase the hit ratio.

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4.21 a. First, 2.5 ns are needed to determine that a cache miss occurs. Then, the required line is read into the cache. Then an additional 2.5 ns are needed to read the requested word.

$$T_{miss} = 2.5 + 50 + (15)(5) + 2.5 = 130 \text{ ns}$$

b. The value T_{miss} from part (a) is equivalent to the quantity (T1 + T2) in Equation (4.1). Under the initial conditions, using Equation (4.1), the average access time is

$$T_s = H \times T_1 + (1 - H) \times (T_1 + T_2) = (0.95)(2.5) + (0.05)(130) = 8.875 \text{ ns}$$

Under the revised scheme, we have:

$$T_{miss} = 2.5 + 50 + (31)(5) + 2.5 = 210 \text{ ns}$$

and

$$T_s = H \times T_1 + (1 - H) \times (T_1 + T_2) = (0.97)(2.5) + (0.03)(210) = 8.725 \text{ ns}$$

4.22 There are three cases to consider:

Location of referenced word	**Probability**	**Total time for access in ns**
In cache	0.9	20
Not in cache, but in main memory	(0.1)(0.6) = 0.06	60 + 20 = 80
Not in cache or main memory	(0.1)(0.4) = 0.04	12ms + 60 + 20 = 12,000,080

So the average access time would be:

Avg = (0.9)(20) + (0.06)(80) + (0.04)(12000080) = 480026 ns

4.23 a. Consider the execution of 100 instructions. Under **write-through**, this creates 200 cache references (168 read references and 32 write references). On average, the read references result in (0.03) $\times$ 168 = 5.04 read misses. For each read miss, a line of memory must be read in, generating 5.04 $\times$ 8 = 40.32 physical words of traffic. For write misses, a single word is written back, generating 32 words of traffic. **Total traffic**: 72.32 words. For **write back**, 100 instructions create 200 cache references and thus 6 cache misses. Assuming 30% of lines are dirty, on average 1.8 of these misses require a line write before a line read. Thus, **total traffic** is (6 + 1.8) $\times$ 8 = 62.4 words. The traffic rate:
Write through = 0.7232 byte/instruction
Write back = 0.624 bytes/instruction

b. For write-through: [(0.05) $\times$ 168 $\times$ 8] + 32 = 99.2 $\rightarrow$ 0.992 bytes/instruction

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For write-back: $(10 + 3) \times 8 = 104 \rightarrow 0.104$ bytes/instruction

c. For write-through: $[(0.07) \times 168 \times 8] + 32 = 126.08 \rightarrow 0.12608$ bytes/instruction
For write-back: $(14 + 4.2) \times 8 = 145.6 \rightarrow 0.1456$ bytes/instruction

d. A 5% miss rate is roughly a crossover point. At that rate, the memory traffic is about equal for the two strategies. For a lower miss rate, write-back is superior. For a higher miss rate, write-through is superior.

4.24 a. One clock cycle equals 60 ns, so a cache access takes 120 ns and a main memory access takes 180 ns. The effective length of a memory cycle is $(0.9 \times 120) + (0.1 \times 180) = 126$ ns.

b. The calculation is now $(0.9 \times 120) + (0.1 \times 300) = 138$ ns. Clearly the performance degrades. However, note that although the memory access time increases by 120 ns, the average access time increases by only 12 ns.

4.25 a. For a 1 MIPS processor, the average instruction takes 1000 ns to fetch and execute. On average, an instruction uses two bus cycles for a total of 600 ns, so the bus utilization is 0.6

b. For only half of the instructions must the bus be used for instruction fetch. Bus utilization is now (150 + 300)/1000 = 0.45. This reduces the waiting time for other bus requestors, such as DMA devices and other microprocessors.

4.26 a. $T_a = T_c + (1 - H)T_b + W(T_m - T_c)$

b. $T_a = T_c + (1 - H)T_b + W_b(1 - H)T_b = T_c + (1 - H)(1 + W_b)T_b$

4.27 $T_a = [T_{c1} + (1 - H_1)T_{c2}] + (1 - H_2)T_m$

4.28 a. miss penalty = 1 + 4 = 5 clock cycles

b. miss penalty = $4 \times (1 + 4)$ = 20 clock cycles

c. miss penalty = miss penalty for one word + 3 = 8 clock cycles.

4.29 The average miss penalty equals the miss penalty times the miss rate. For a line size of one word, average miss penalty = 0.032 x 5 = 0.16 clock cycles. For a line size of 4 words and the nonburst transfer, average miss penalty = 0.011 x 20 = 0.22 clock cycles. For a line size of 4 words and the burst transfer, average miss penalty = 0.011 x 8 = 0.132 clock cycles.

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CHAPTER 5 INTERNAL MEMORY

ANSWERS TO QUESTIONS

5.1 They exhibit two stable (or semistable) states, which can be used to represent binary 1 and 0; they are capable of being written into (at least once), to set the state; they are capable of being read to sense the state.

5.2 **(1)** A memory in which individual words of memory are directly accessed through wired-in addressing logic. **(2)** Semiconductor main memory in which it is possible both to read data from the memory and to write new data into the memory easily and rapidly.

5.3 SRAM is used for cache memory (both on and off chip), and DRAM is used for main memory.

5.4 SRAMs generally have faster access times than DRAMs. DRAMS are less expensive and smaller than SRAMs.

5.5 A **DRAM cell** is essentially an analog device using a capacitor; the capacitor can store any charge value within a range; a threshold value determines whether the charge is interpreted as 1 or 0. A **SRAM cell** is a digital device, in which binary values are stored using traditional flip-flop logic-gate configurations.

5.6 Microprogrammed control unit memory; library subroutines for frequently wanted functions; system programs; function tables.

5.7 **EPROM** is read and written electrically; before a write operation, all the storage cells must be erased to the same initial state by exposure of the packaged chip to ultraviolet radiation. Erasure is performed by shining an intense ultraviolet light through a window that is designed into the memory chip. **EEPROM** is a read-mostly memory that can be written into at any time without erasing prior contents; only the byte or bytes addressed are updated. **Flash memory** is intermediate between EPROM and EEPROM in both cost and functionality. Like EEPROM, flash memory uses an electrical erasing technology. An entire flash memory can be erased in one or a few seconds, which is much faster than EPROM. In addition, it is possible to erase just blocks of memory rather than an

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entire chip. However, flash memory does not provide byte-level erasure. Like EPROM, flash memory uses only one transistor per bit, and so achieves the high density (compared with EEPROM) of EPROM.

5.8 A0 - A1 = address lines:. CAS = column address select:. D1 - D4 = data lines. NC: = no connect. OE: output enable. RAS = row address select:. Vcc: = voltage source. Vss: = ground. WE: write enable.

5.9 A bit appended to an array of binary digits to make the sum of all the binary digits, including the parity bit, always odd (odd parity) or always even (even parity).

5.10 A syndrome is created by the XOR of the code in a word with a calculated version of that code. Each bit of the syndrome is 0 or 1 according to if there is or is not a match in that bit position for the two inputs. If the syndrome contains all 0s, no error has been detected. If the syndrome contains one and only one bit set to 1, then an error has occurred in one of the 4 check bits. No correction is needed. If the syndrome contains more than one bit set to 1, then the numerical value of the syndrome indicates the position of the data bit in error. This data bit is inverted for correction.

5.11 Unlike the traditional DRAM, which is asynchronous, the SDRAM exchanges data with the processor synchronized to an external clock signal and running at the full speed of the processor/memory bus without imposing wait states.

5.12 DDR RAM) provides several features that dramatically increase the data rate over ordinary DRAM. DDR achieves higher data rates in three ways. First, the data transfer is synchronized to both the rising and falling edge of the clock, rather than just the rising edge. This doubles the data rate; hence the term *double data rate*. Second, DDR uses higher clock rate on the bus to increase the transfer rate. Third, a buffering scheme is used.

5.13 In **NOR flash memory**, the basic unit of access is a bit, referred to as a *memory cell*. Cells in NOR flash are connected in parallel to the bit lines so that each cell can be read/write/erased individually. If any memory cell of the device is turned on by the corresponding word line, the bit line goes low. This is similar in function to a NOR logic gate. **NAND flash memory** is organized in transistor arrays with 16 or 32 transistors in series. The bit line goes low only if all the transistors in the corresponding word lines are turned on. This is similar in function to a NAND logic gate.

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5.14 Spin-transfer torque RAM (STT-RAM) is a type of magnetic RAM with a new write mechanism, in which the magnetization of the free layer is flipped by the electrical current directly.
Phase-change RAM (PCRAM) is based on a chalcogenide alloy material, which is similar to those commonly used in optical storage media (compact discs and digital versatile discs). The data storage capability is achieved from the resistance differences between an amorphous (high-resistance) and a crystalline (low-resistance) phase of the chalcogenide-based material.
Resistive RAM (ReRAM) works by creating resistance rather than directly storing charge.

ANSWERS TO PROBLEMS

5.1 The 1-bit-per-chip organization has several advantages. It requires fewer pins on the package (only one data out line); therefore, a higher density of bits can be achieved for a given size package. Also, it is somewhat more reliable because it has only one output driver. These benefits have led to the traditional use of 1-bit-per-chip for RAM. In most cases, ROMs are much smaller than RAMs and it is often possible to get an entire ROM on one or two chips if a multiple-bits-per-chip organization is used. This saves on cost and is sufficient reason to adopt that organization.

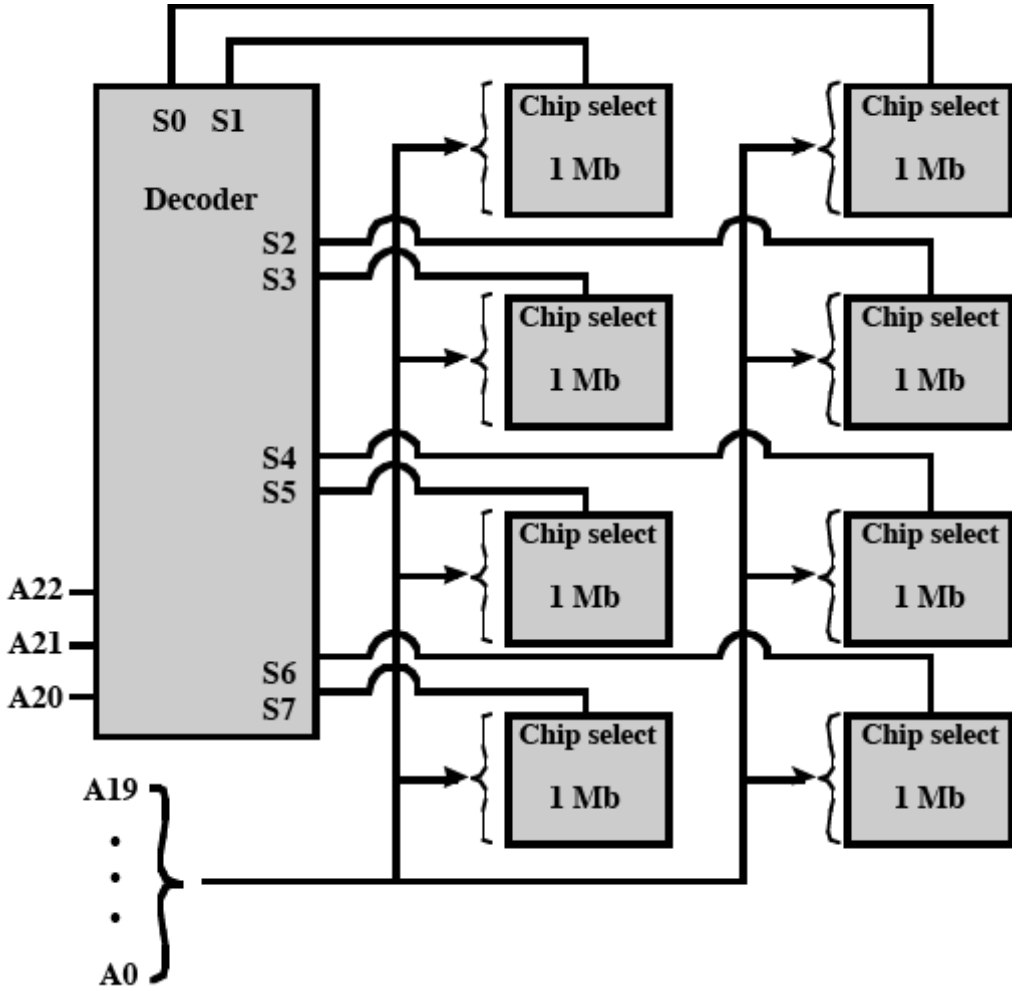
5.2 In 1 ms, the time devoted to refresh is 64×150 ns = 9600 ns. The fraction of time devoted to memory refresh is $(9.6 \times 10^{-6}\ \text{s})/10^{-3}\ \text{s} = 0.0096$, which is approximately 1%.

5.3 **a.** Memory cycle time = 60 + 40 = 100 ns. The maximum data rate is 1 bit every 100 ns, which is 10 Mbps.
b. 320 Mbps = 40 MB/s.

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5.4



5.5 **a.** The length of a clock cycle is 100 ns. Mark the beginning of T_1 as time 0.Address Enable returns to a low at 75. [ ] goes active 50 ns later, or time 125. Data must become available by the DRAMs at time 300 – 60 = 240. Hence, access time must be no more than 240 – 125 = 115 ns.

b. A single wait state will increase the access time requirement to 115 + 100 = 215 ns. This can easily be met by DRAMs with access times of 150 ns.

5.6 **a.** The refresh period from row to row must be no greater than 4000/256 = 15.625 μs.

b. An 8-bit counter is needed to count 256 rows (2^8 = 256).

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5.7 a.

pulse a = write
pulse b = write
pulse c = write
pulse d = write
pulse e= write
pulse f = write
pulse g = store-disable outputs
pulse h = read
pulse i = read
pulse j = read
pulse k = read
pulse l = read
pulse m = read
pulse n = store-disable outputs

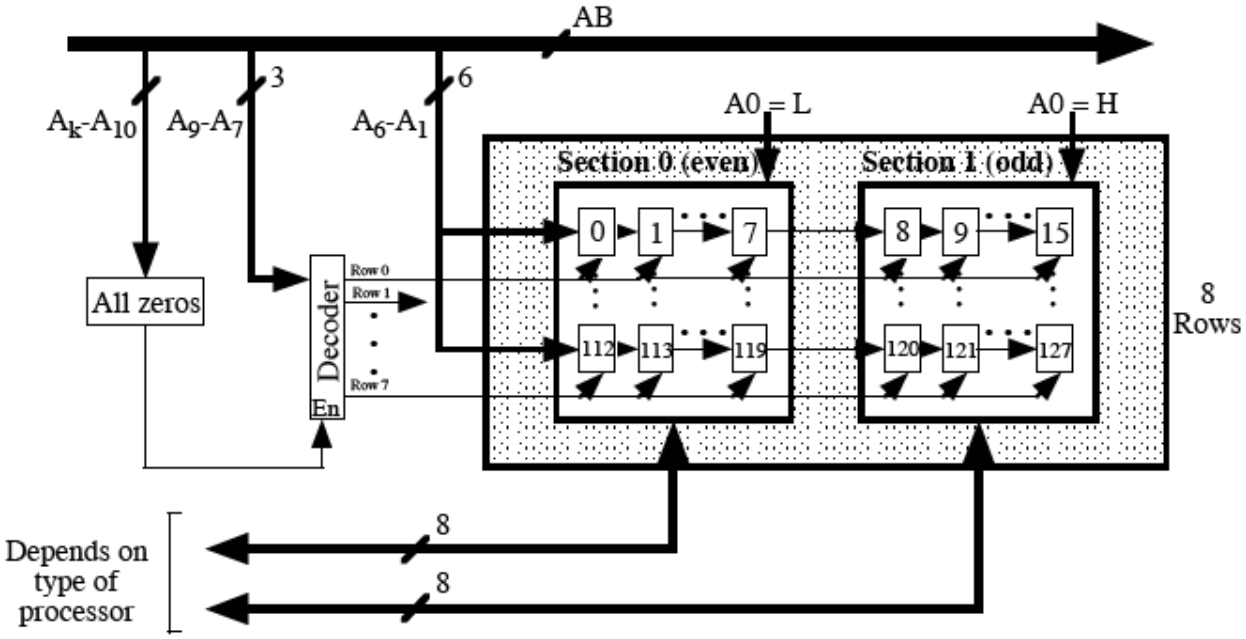
b. Data is read in via pins (D3, D2, D1, D0)
word 0 = 1111 (written into location 0 during pulse a)
word 1 = 1110 (written into location 0 during pulse b)
word 2 = 1101 (written into location 0 during pulse c)
word 3 = 1100 (written into location 0 during pulse d)
word 4 = 1011 (written into location 0 during pulse e)
word 5 = 1010 (written into location 0 during pulse f)
word 6 = random (did not write into this location 0)

c. Output leads are (O3, O2, O1, O0)
pulse h: 1111 (read location 0)
pulse i: 1110 (read location 1)
pulse j: 1101 (read location 2)
pulse k: 1100 (read location 3)
pulse l: 1011 (read location 4)
pulse m: 1010 (read location 5)

5.8 8192/64 = 128 chips; arranged in 8 rows by 64 columns:

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5.9 Total memory is 1 megabyte = 8 megabits. It will take 32 DRAMs to construct the memory (32 $\times$ 256 Kb = 8 Mb). The composite failure rate is 2000 $\times$ 32 = 64,000 FITS. From this, we get a MTBF = 10^9/64,000 = 15625 hours = 22 months.

5.10 The stored word is 001101001111, as shown in Figure 5.10. Now suppose that the only error is in C8, so that the fetched word is 001111001111. Then the received block results in the following table:

Position	12	11	10	9	8	7	6	5	4	3	2	1
Bits	D8	D7	D6	D5	C8	D4	D3	D2	C4	D1	C2	C1
Block	0	0	1	1	1	1	0	0	1	1	1	1
Codes			1010	1001		0111				0011		

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The check bit calculation after reception:

Position	Code
Hamming	1111
10	1010
9	1001
7	0111
3	0011
XOR = syndrome	1000

The nonzero result detects and error and indicates that the error is in bit position 8, which is check bit C8.

5.11 Data bits with value 1 are in bit positions 12, 11, 5, 4, 2, and 1:

Position	12	11	10	9	8	7	6	5	4	3	2	1
Bits	D8	D7	D6	D5	C8	D4	D3	D2	C4	D1	C2	C1
Block	1	1	0	0		0	0	1		0		
Codes	1100	1011						0101				

The check bits are in bit numbers 8, 4, 2, and 1.
Check bit 8 calculated by values in bit numbers: 12, 11, 10 and 9
Check bit 4 calculated by values in bit numbers: 12, 7, 6, and 5
Check bit 2 calculated by values in bit numbers: 11, 10, 7, 6 and 3
Check bit 1 calculated by values in bit numbers: 11, 9, 7, 5 and 3
Thus, the check bits are: 0 0 1 0

5.12 The Hamming Word initially calculated was:

bit number:

12	11	10	9	8	7	6	5	4	3	2	1
0	0	1	1	0	1	0	0	1	1	1	1

Doing an exclusive-OR of 0111 and 1101 yields 1010 indicating an error in bit 10 of the Hamming Word. Thus, the data word read from memory was 00011001.

5.13 Need K check bits such that $1024 + K \leq 2^K - 1$.
The minimum value of K that satisfies this condition is 11.

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5.14 As Table 5.2 indicates, 5 check bits are needed for an SEC code for 16-bit data words. The layout of data bits and check bits:

Bit Position	Position Number	Check Bits	Data Bits
21	10101		M16
20	10100		M15
19	10011		M14
18	10010		M13
17	10001		M12
16	10000	C16	
15	01111		M11
14	01110		M10
13	01101		M9
12	01100		M8
11	01011		M7
10	01010		M6
9	01001		M5
8	01000	C8	
7	00111		M4
6	00110		M3
5	00101		M2
4	00100	C4	
3	00011		M1
2	00010	C2	
1	00001	C1	

The equations are calculated as before, for example,
C1= M1 ⊕ M2 ⊕M4 ⊕ M5 ⊕ M7 ⊕ M9 ⊕ M11 ⊕ M12 ⊕ M14 ⊕ M16.

For the word 0101000000111001, the code is
C16 = 1; C8 = 1; C 4 = 1; C2 = 1; C1 = 0.

If an error occurs in data bit 4:
C16 = 1 ; C8 =1; C4 = 0; C2 = 0; C1 = 1.

Comparing the two:

C16	C8	C4	C2	C1
1	1	1	1	0
1	1	0	0	1
0	0	1	1	1

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The result is an error identified in bit position 7, which is data bit 4.

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CHAPTER 6 EXTERNAL MEMORY

ANSWERS TO QUESTIONS

6.1 Improvement in the uniformity of the magnetic film surface to increase disk reliability. A significant reduction in overall surface defects to help reduce read/write errors. Ability to support lower fly heights (described subsequently). Better stiffness to reduce disk dynamics. Greater ability to withstand shock and damage

6.2 The write mechanism is based on the fact that electricity flowing through a coil produces a magnetic field. Pulses are sent to the write head, and magnetic patterns are recorded on the surface below, with different patterns for positive and negative currents. An electric current in the wire induces a magnetic field across the gap, which in turn magnetizes a small area of the recording medium. Reversing the direction of the current reverses the direction of the magnetization on the recording medium.

6.3 The read head consists of a partially shielded magnetoresistive (MR) sensor. The MR material has an electrical resistance that depends on the direction of the magnetization of the medium moving under it. By passing a current through the MR sensor, resistance changes are detected as voltage signals.

6.4 For the **constant angular velocity** (CAV) system, the number of bits per track is constant. An increase in density is achieved with **multiple zoned recording**, in which the surface is divided into a number of zones, with zones farther from the center containing more bits than zones closer to the center.

6.5 On a magnetic disk. data is organized on the platter in a concentric set of rings, called **tracks**. Data are transferred to and from the disk in **sectors**. For a disk with multiple platters, the set of all the tracks in the same relative position on the platter is referred to as a **cylinder**.

6.6 512 bytes.

6.7 On a movable-head system, the time it takes to position the head at the track is known as **seek time**. Once the track is selected, the disk

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controller waits until the appropriate sector rotates to line up with the head. The time it takes for the beginning of the sector to reach the head is known as **rotational delay**. The sum of the seek time, if any, and the rotational delay equals the **access time**, which is the time it takes to get into position to read or write. Once the head is in position, the read or write operation is then performed as the sector moves under the head; this is the data transfer portion of the operation and the time for the transfer is the **transfer time**.

6.8 **1.** RAID is a set of physical disk drives viewed by the operating system as a single logical drive. **2.** Data are distributed across the physical drives of an array. **3.** Redundant disk capacity is used to store parity information, which guarantees data recoverability in case of a disk failure.

6.9 **0:** Non-redundant **1:** Mirrored; every disk has a mirror disk containing the same data. **2:** Redundant via Hamming code; an error-correcting code is calculated across corresponding bits on each data disk, and the bits of the code are stored in the corresponding bit positions on multiple parity disks. **3:** Bit-interleaved parity; similar to level 2 but instead of an error-correcting code, a simple parity bit is computed for the set of individual bits in the same position on all of the data disks. **4:** Block-interleaved parity; a bit-by-bit parity strip is calculated across corresponding strips on each data disk, and the parity bits are stored in the corresponding strip on the parity disk. **5:** Block-interleaved distributed parity; similar to level 4 but distributes the parity strips across all disks. **6:** Block-interleaved dual distributed parity; two different parity calculations are carried out and stored in separate blocks on different disks.

6.10 The disk is divided into strips; these strips may be physical blocks, sectors, or some other unit. The strips are mapped round robin to consecutive array members. A set of logically consecutive strips that maps exactly one strip to each array member is referred to as a *stripe*.

6.11 For RAID level 1, redundancy is achieved by having two identical copies of all data. For higher levels, redundancy is achieved by the use of error-correcting codes.

6.12 In a **parallel access** array, all member disks participate in the execution of every I/O request. Typically, the spindles of the individual drives are synchronized so that each disk head is in the same position on each disk at any given time. In an **independent access** array, each member disk operates independently, so that separate I/O requests can be satisfied in parallel.

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6.13 For the **constant angular velocity (CAV)** system, the number of bits per track is constant. At a **constant linear velocity (CLV)**, the disk rotates more slowly for accesses near the outer edge than for those near the center. Thus, the capacity of a track and the rotational delay both increase for positions nearer the outer edge of the disk.

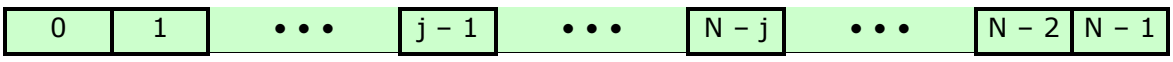
6.14 **1.** Bits are packed more closely on a DVD. The spacing between loops of a spiral on a CD is 1.6 μm and the minimum distance between pits along the spiral is 0.834 μm. The DVD uses a laser with shorter wavelength and achieves a loop spacing of 0.74 μm and a minimum distance between pits of 0.4 μm. The result of these two improvements is about a seven-fold increase in capacity, to about 4.7 GB. **2.** The DVD employs a second layer of pits and lands on top of the first layer A dual-layer DVD has a semireflective layer on top of the reflective layer, and by adjusting focus, the lasers in DVD drives can read each layer separately. This technique almost doubles the capacity of the disk, to about 8.5 GB. The lower reflectivity of the second layer limits its storage capacity so that a full doubling is not achieved. **3.** The DVD-ROM can be two sided whereas data is recorded on only one side of a CD. This brings total capacity up to 17 GB.

6.15 The typical recording technique used in serial tapes is referred to as **serpentine recording**. In this technique, when data are being recorded, the first set of bits is recorded along the whole length of the tape. When the end of the tape is reached, the heads are repositioned to record a new track, and the tape is again recorded on its whole length, this time in the opposite direction. That process continues, back and forth, until the tape is full.

ANSWERS TO PROBLEMS

6.1 As stated in the discussion preceding Equation (6.1) the time to access a block of data is the sum of the seek time and the average rotational delay, which accounts for the first two terms in the equation. Once the block is accessed, the third term expresses the transfer time. Thus, access time plus transfer time equals the total time to read or write a block of data.

6.2 It will be useful to keep the following representation of the N tracks of a disk in mind:



a. Let us use the notation Ps [j/t] = Pr [seek of length j when head is currently positioned over track t]. Recognize that each of the N

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tracks is equally likely to be requested. Therefore the unconditional probability of selecting any particular track is 1/N. We can then state:

$$\text{Ps}[j/t] = \frac{1}{N} \quad \text{if} \quad t \le j - 1 \ \text{ OR } \ t \ge N - j$$

$$\text{Ps}[j/t] = \frac{2}{N} \quad \text{if} \quad j - 1 < t < N - j$$

In the former case, the current track is so close to one end of the disk (track 0 or track N – 1) that only one track is exactly j tracks away. In the second case, there are two tracks that are exactly j tracks away from track t, and therefore the probability of a seek of length j is the probability that either of these two tracks is selected, which is just 2/N.

b. Let Ps [K] = Pr [seek of length K, independent of current track position]. Then:

$$\text{Ps}[K] = \sum_{t=0}^{N-1} \text{Ps}[K/t] \times \text{Pr}[\text{current track is track t}]$$

$$= \frac{1}{N} \sum_{t=0}^{N-1} \text{Ps}[K/t]$$

From part (a), we know that Ps[K/t] takes on the value 1/N for 2K of the tracks, and the value 2/N for (N – 2K) of the tracks. So

$$\text{Ps}[K] = \frac{1}{N}\left[\frac{2K}{N} + \frac{2(N-2K)}{N}\right] = \frac{2K + 2(N-2K)}{N^2} = \frac{2}{N} - \frac{2K}{N^2}$$

c.

$$\text{E}[K] = \sum_{K=0}^{N-1} K \times \text{Ps}[K] = \sum_{K=0}^{N-1} \frac{2K}{N} - \frac{2K^2}{N^2} = \frac{2}{N}\sum_{K=0}^{N-1} K - \frac{2}{N^2}\sum_{K=0}^{N-1} K^2$$

$$= \frac{2}{N}\frac{(N-1)N}{2} - \frac{2}{N^2}\frac{(N-1)N(2N-1)}{6} = (N-1) - \frac{(N-1)(2N-1)}{3N}$$

$$= \frac{3N(N-1) - (N-1)(2N-1)}{3N} = \frac{N^2-1}{3N}$$

d. This follows directly from the last equation.

6.3 $$t_{sector} = t_S + \frac{1}{2r} + \frac{n}{rN}$$

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6.4 **a.** Capacity = $8 \times 512 \times 64 \times 1$ kB = 256 MB

b. Rotational latency = rotation_time/2 = $60/(3600 \times 2)$ = 8.3 ms. Average access time = seek time + rotational latency = 16.3 ms

c. Each cylinder consists of 8 tracks $\times$ 64 sectors/track $\times$ 1 kB/sector = 512 kB, so 5 MB requires exactly 10 cylinders. The disk will need the seek time of 8 ms to find cylinder *i*, 8.3 ms on average to find sector 0, and $8 \times (60/3.6)$ = 133.3 ms to read all 8 tracks on one cylinder. Then, the time needed to move to the next adjoining cylinder is 1.5 ms, which is the track-to-track access time. Assume a rotational latency before each track.

Access Time = $8 + 9 \times (8.3 + 133.3 + 1.5) + (8.3 + 133.3)$ = 1425.5 ms

d.

$$\text{Burst rate} = \frac{\text{revolutions}}{\text{second}} \times \frac{\text{sectors}}{\text{revolution}} \times \frac{\text{bytes}}{\text{secter}} = \frac{3600}{60} \times 64 \times 1 \text{ KB} = 3.84 \text{ MB/s}$$

6.5 **a.** If we assume that the head starts at track 0, then the calculations are simplified. If the request track is track 0, then the seek time is 0; if the requested track is track 29,999, then the seek time is the time to traverse 29,999 tracks. For a random request, on average the number of tracks traversed is 29,999/2 = 14999.5 tracks. At one ms per 100 tracks, the average seek time is therefore 149.995 ms.

b. At 7200 rpm, there is one revolution every 8.333 ms. Therefore, the average rotational delay is 4.167 ms.

c. With 600 sectors per track and the time for one complete revolution of 8.333 ms, the transfer time for one sector is 8.333 ms/600 = 0.01389 ms.

d. The result is the sum of the preceding quantities, or approximately 154 ms.

6.6 Each sector can hold 4 logical records. The required number of sectors is 300,000/4 = 75,000 sectors. This requires 75,000/96 = 782 tracks, which in turn requires 782/110 = 8 surfaces.

6.7 **a.** The time consists of the following components: sector read time; track access time; rotational delay; and sector write time. The time to read or write 1 sector is calculated as follows: A single revolution to read or write an entire track takes 60,000/360 = 16.7 ms. Time to read or write a single sector = 16.7/32 = 0.52 ms. Track access time = 2 ms, because the head moves between adjacent tracks. The rotational delay is the time required for the head to line up with sector 1 again. This is $16.7 \times (31/32)$ = 16.2 ms. The head movement time of 2 ms overlaps the with the 16.2 ms of rotational

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delay, and so only the rotational delay time is counted. Total transfer time = 0.52 + 16.2 + 0.52 = 17.24 ms.

b. The time to read or write an entire track is simply the time for a single revolution, which is 16.7 ms. Between the read and the write there is a head movement time of 2 ms to move from track 8 to track 9. During this time the head moves past 3 sectors and most of a fourth sector. However, because the entire track is buffered, sectors can be written back in a different sequence from the read sequence. Thus, the write can start with sector 5 of track 9. This sector is reached $0.52 \times 4 = 2.08$ ms after the completion of the read operation. Thus the total transfer time = 16.7 + 2.08 + 16.7 = 35.48 ms.

6.8 It depends on the nature of the I/O request pattern. On one extreme, if only a single process is doing I/O and is only doing one large I/O at a time, then disk striping improves performance. If there are many processes making many small I/O requests, then a nonstriped array of disks should give comparable performance to RAID 0.

6.9

RAID 0: 800 GB	RAID 4: 600 GB
RAID 1: 400 GB	RAID 5: 600 GB
RAID 3: 600 GB	RAID 6: 400 GB

6.10 With the CD scheme, bit density improves by a factor of 3, but there is an increase in the number of bits by a factor of 14/8. Net improvement factor is $3 \times (8/14) \approx 1.7$. That is, the CD scheme has a data storage density 1.7 times greater than the direct recording scheme.

6.11 **a.** $2 \times 3 \times \$150 = \$900$

b. $\$2500 + (3 \times 3 \times \$50) = \$2950$

c. Let Z = the number of GB at which the two approaches yield approximately the same cost. For the disk, the cost is $C_d = (Z/500) \times 3 \times \$150$. For tape, the cost is $C_t = 2500 + ((Z/400) \times 3 \times \$50)$. If we set Cd = Ct and solve for Z, we get Z = 4762. So the size of the backup would have to be about 5 TB for tape to be less expensive.

d. One where you keep a lot of backup sets.

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CHAPTER 7 INPUT/OUTPUT

ANSWERS TO QUESTIONS

7.1 **Human readable:** Suitable for communicating with the computer user. **Machine readable:** Suitable for communicating with equipment. **Communication:** Suitable for communicating with remote devices

7.2 The most commonly used text code is the International Reference Alphabet (IRA), in which each character is represented by a unique 7-bit binary code; thus, 128 different characters can be represented.

7.3 Control and timing. Processor communication. Device communication. Data buffering. Error detection.

7.4 **Programmed I/O:** The processor issues an I/O command, on behalf of a process, to an I/O module; that process then busy-waits for the operation to be completed before proceeding. **Interrupt-driven I/O:** The processor issues an I/O command on behalf of a process, continues to execute subsequent instructions, and is interrupted by the I/O module when the latter has completed its work. The subsequent instructions may be in the same process, if it is not necessary for that process to wait for the completion of the I/O. Otherwise, the process is suspended pending the interrupt and other work is performed. **Direct memory access (DMA):** A DMA module controls the exchange of data between main memory and an I/O module. The processor sends a request for the transfer of a block of data to the DMA module and is interrupted only after the entire block has been transferred.

7.5 With **memory-mapped I/O**, there is a single address space for memory locations and I/O devices. The processor treats the status and data registers of I/O modules as memory locations and uses the same machine instructions to access both memory and I/O devices. With **isolated I/O**, a command specifies whether the address refers to a memory location or an I/O device. The full range of addresses may be available for both.

7.6 Four general categories of techniques are in common use: multiple interrupt lines; software poll; daisy chain (hardware poll, vectored); bus arbitration (vectored).

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7.7 The processor pauses for each bus cycle stolen by the DMA module.

ANSWERS TO PROBLEMS

7.1 In the first addressing mode, $2^8 = 256$ ports can be addressed. Typically, this would allow 128 devices to be addressed. However, an opcode specifies either an input or output operation, so it is possible to reuse the addresses, so that there are 256 input port addresses and 256 output port addresses. In the second addressing mode, $2^{16} = 64K$ port addresses are possible.

7.2 In direct addressing mode, an instruction can address up to $2^{16} = 64K$ ports. In indirect addressing mode, the port address resides in a 16-bit registers, so again, the instruction can address up to $2^{16} = 64K$ ports.

7.3 64 kB

7.4 Using non-block I/O instructions, the transfer takes $20 \times 128 = 2560$ clock cycles. With block I/O, the transfer takes $5 \times 128 = 640$ clock cycles (ignoring the one-time fetching of the iterative instruction and its operands). The speedup is
$(2560 – 640)/2560 = 0.75$, or 75%.

7.5 **a.** Each I/O device requires one output (from the point of view of the processor) port for commands and one input port for status.
b. The first device requires only one port for data, while the second devices requires and input data port and an output data port. Because each device requires one command and one status port, the total number of ports is seven.
c. seven.

7.6 **a.** The printing rate is slowed to 5 cps.
b. The situation must be treated differently with input devices such as the keyboard. It is necessary to scan the buffer at a rate of at least once per 60 ms. Otherwise, there is the risk of overwriting characters in the buffer.

7.7 At 8 MHz, the processor has a clock period of 0.125 μs, so that an instruction cycle takes $12 \times 0.125 = 1.5$ μs. To check status requires one input-type instruction to read the device status register, plus at least one other instruction to examine the register contents. If the device is ready, one output-type instruction is needed to present data to the device handler. The total is 3 instructions, requiring 4.5 μs.

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7.8 Advantages of memory mapped I/O:

1. No additional control lines are needed on the bus to distinguish memory commands from I/O commands.
2. Addressing is more flexible. Examples: The various addressing modes of the instruction set can be used, and various registers can be used to exchange data with I/O modules.

Disadvantages of memory-mapped I/O:

1. Memory-mapped I/O uses memory-reference instructions, which in most machines are longer than I/O instructions. The length of the program therefore is longer.
2. The hardware addressing logic to the I/O module is more complex, because the device address is longer.

7.9 a. The processor scans the keyboard 10 times per second. In 8 hours, the number of times the keyboard is scanned is $10 \times 60 \times 60 \times 8 = 288{,}000$.

b. Only 60 visits would be required. The reduction is $1 - (60/288000) = 0.999$, or 99.9%

7.10 1001 0000

7.11 a. The device generates 8000 interrupts per second or a rate of one every 125 μs. If each interrupt consumes 100 μs, then the fraction of processor time consumed is 100/125 = 0.8

b. In this case, the time interval between interrupts is $16 \times 125 = 2000$ μs. Each interrupt now requires 100 μs for the first character plus the time for transferring each remaining character, which adds up to $8 \times 15 = 120$ μs, for a total of 220 μs. The fraction of processor time consumed is 220/2000 = 0.11

c. The time per byte has been reduced by 6 μs, so the total time reduction is $16 \times 6 = 96$ μs. The fraction of processor time consumed is therefore (220 – 96)/2000 = 0.062. This is an improvement of almost a factor of 2 over the result from part (b).

7.12 If a processor is held up in attempting to read or write memory, usually no damage occurs except a slight loss of time. However, a DMA transfer may be to or from a device that is receiving or sending data in a stream (e.g., disk or tape), and cannot be stopped. Thus, if the DMA module is held up (denied continuing access to main memory), data will be lost.

7.13 Let us ignore data read/write operations and assume the processor only fetches instructions. Then the processor needs access to main memory once every microsecond. The DMA module is transferring characters at a rate of 1200 characters per second, or one every 833

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μs. The DMA therefore "steals" every 833rd cycle. This slows down the processor approximately $\frac{1}{833} \times 100\% = 0.12\%$

7.14 **a.** For the actual transfer, the time needed is (128 bytes)/(50 kBps) = 2.56 ms. Added to this is the time to transfer bus control at the beginning and end of the transfer, which is 250 + 250 = 500 ns. This additional time is negligible, so that the transfer time can be considered as 2.56 ms.

b. The time to transfer one byte in cycle stealing mode is 250 + 500 + 250 = 1000 ns = 1 μs. Total amount of time the bus is occupied for the transfer is 128 μs. This is less than the result from part (a) by a factor of 20.

7.15 **a.** At 5 MHz, on clock cycle takes 0.2 μs. A transfer of one byte therefore takes 0.6 μs.

b. The data rate is $1/(0.6 \times 10^{-6})$ = 1.67 MB/s

c. Two wait states add an addition 0.4 μs, so that a transfer of one byte takes 1 μs. The resulting data rate is 1 MB/s.

7.16 A DMA cycle could take as long as 0.75 μs without the need for wait states. This corresponds to a clock period of 0.75/3 = 0.25 μs, which in turn corresponds to a clock rate of 4 MHz. This approach would eliminate the circuitry associated with wait state insertion and also reduce power dissipation.

7.17 **a.** Telecommunications links can operate continuously, so burst mode cannot be used, as this would tie up the bus continuously. Cycle-stealing is needed.

b. Because all 4 links have the same data rate, they should be given the same priority.

7.18 Only one device at a time can be serviced on a selector channel. Thus, Maximum rate = $800 + 800 + 2 \times 6.6 + 2 \times 1.2 + 10 \times 1 = 1625.6$ kB/sec

7.19 **a.** The processor can only devote 5% of its time to I/O. Thus the maximum I/O instruction execution rate is $10^6 \times 0.05$ = 50,000 instructions per second. The I/O transfer rate is therefore 25,000 words/second.

b. The number of machine cycles available for DMA control is

$10^6(0.05 \times 5 + 0.95 \times 2) = 2.15 \times 10^6$

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If we assume that the DMA module can use all of these cycles, and ignore any setup or status-checking time, then this value is the maximum I/O transfer rate.

7.20 For each case, compute the fraction g of transmitted bits that are data bits. Then the maximum effective data rate ER is

$$ER = gR$$

a. There are 7 data bits, 1 start bit, 1.5 stop bits, and 1 parity bit.

$$g = \frac{7}{1 + 7 + 1 + 1.5} = 7/10.5$$

$$ER = 0.67 \times R$$

b. Each frame contains 48 + 128 = 176 bits. The number of characters is 128/8 = 16, and the number of data bits is $16 \times 7 = 112$.

$$ER = \frac{112}{176} \times R = 0.64 \times R$$

c. Each frame contains 48 = 1024 bits. The number of characters is 1024/8 = 128, and the number of data bits is $128 \times 7 = 896$.

$$ER = \frac{896}{1072} \times R = 0.84 \times R$$

d. With 9 control characters and 16 information characters, each frame contains $(9 + 16) \times 8 = 200$ bits. The number of data bits is $16 \times 7 = 112$ bits.

$$ER = \frac{112}{200} \times R = 0.56 \times R$$

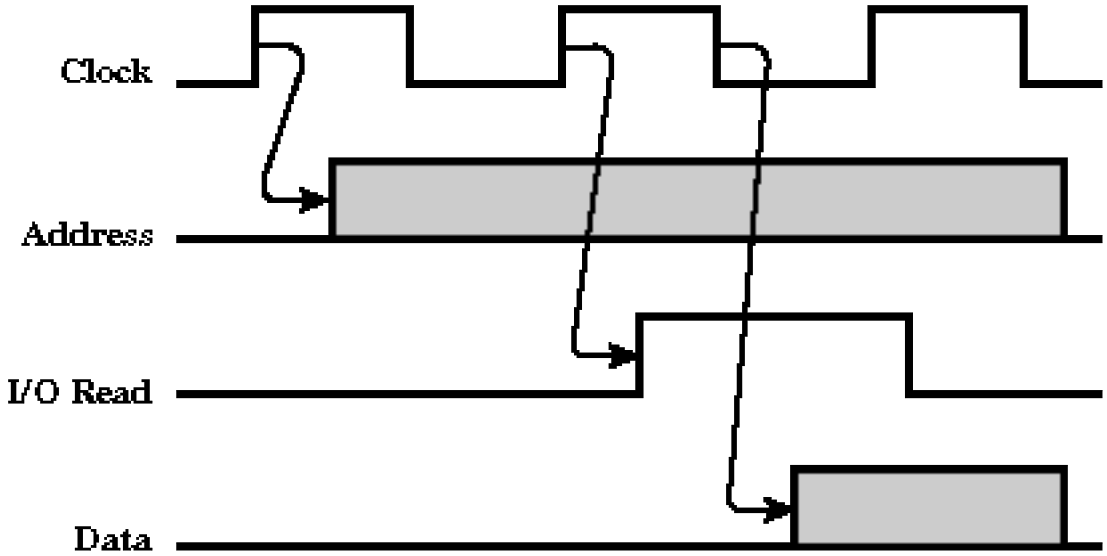
e. With 9 control characters and 128 information characters, each frame contains
$(9 + 128) \times 8 = 1096$ bits. The number of data bits is $128 \times 7 = 896$ bits.

$$ER = \frac{896}{1096} \times R = 0.82 \times R$$

7.21 **a.** Assume that the women are working, or sleeping, or otherwise engaged. The first time the alarm goes off, it alerts both that it is time to work on apples. The next alarm signal causes apple-server

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to pick up an apple an throw it over the fence. The third alarm is a signal to Apple-eater that he can pick up and eat the apple. The transfer of apples is in strict synchronization with the alarm clock, which should be set to exactly match Apple-eater's needs. This procedure is analogous to standard synchronous transfer of data between a device and a computer. It can be compared to an I/O read operation on a typical bus-based system. The timing diagram is as follows:



On the first clock signal, the port address is output to the address bus. On the second signal, the I/O Read line is activated, causing the selected port to place its data on the data bus. On the third clock signal, the CPU reads the data.

A potential problem with synchronous I/O will occur if Apple-eater's needs change. If he must eat at a slower or faster rate than the clock rate, he will either have too many apples or too few.

b. The women agree that Apple-server will pick and throw over an apple whenever he sees Apple-eater's flag waving. One problem with this approach is that if Apple-eater leaves his flag up, Apple-server will see it all the time and will inundate her friend with apples. This problem can be avoided by giving Apple-server a flag and providing for the following sequence:

1. Apple-eater raises her "hungry" flag when ready for an apple.
2. Apple-server sees the flag and tosses over an apple.
3. Apple-server briefly waves her "apple-sent" flag
4. Apple-eater sees the "apple-sent" flag, takes down her "hungry" flag, and grabs the apple.
5. Apple-eater keeps her "hungry" flag stays down until she needs another apple.

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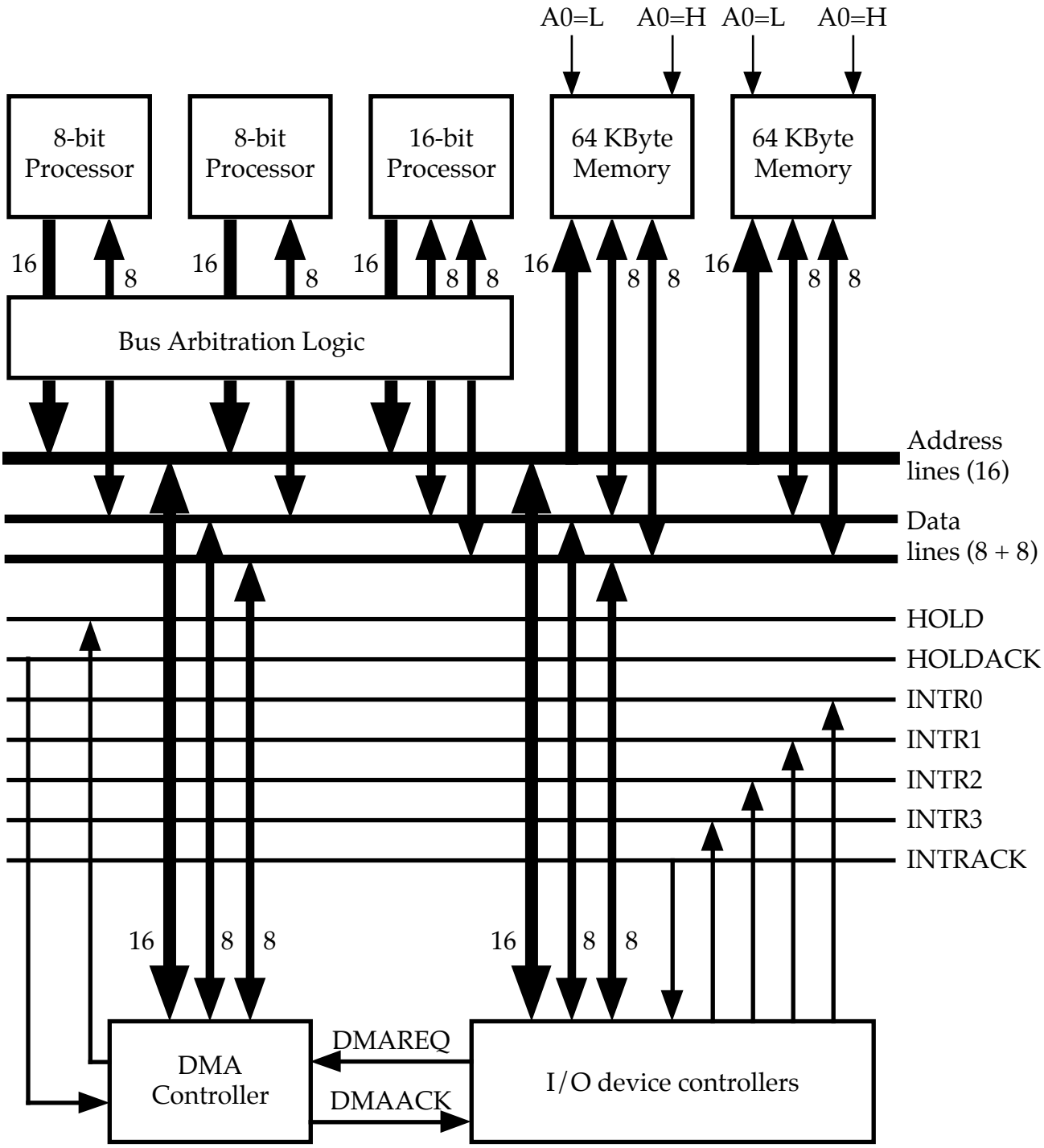
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This procedure is analogous to asynchronous I/O. Unfortunately, Apple-server may be doing something other than watching for her friend's flag (like sleeping!). In that case, she will not see the flag, and Apple-eater will go hungry. One solution is to not permit apple-server to do anything but look for her friend's flag. This is a polling, or wait-loop, approach, which is clearly inefficient.

c. Assume that the string that goes over the fence and is tied to Apple-server's wrist. Apple-eater can pull the string when she needs an apple. When Apple-server feels a tug on the string, she stops what she is doing and throws over an apple. The string corresponds to an interrupt signal and allows Apple-server to use her time more efficiently. Moreover, if Apple-server is doing something really important, she can temporarily untie the string, disabling the interrupt.

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CHAPTER 8 OPERATING SYSTEM SUPPORT

ANSWERS TO QUESTIONS

8.1 The operating system (OS) is the software that controls the execution of programs on a processor and that manages the processor's resources.

8.2 **Program creation:** The operating system provides a variety of facilities and services, such as editors and debuggers, to assist the programmer in creating programs. **Program execution:** A number of tasks need to be performed to execute a program. Instructions and data must be loaded into main memory, I/O devices and files must be initialized, and other resources must be prepared. **Access to I/O devices:** Each I/O device requires its own peculiar set of instructions or control signals for operation. **Controlled access to files:** In the case of files, control must include an understanding of not only the nature of the I/O device (disk drive, tape drive) but also the file format on the storage medium. **System access:** In the case of a shared or public system, the operating system controls access to the system as a whole and to specific system resources. **Error detection and response:** A variety of errors can occur while a computer system is running. **Accounting:** A good operating system will collect usage statistics for various resources and monitor performance parameters such as response time.

8.3 **Long-term scheduling:** The decision to add to the pool of processes to be executed. **Medium-term scheduling:** The decision to add to the number of processes that are partially or fully in main memory. **Short-term scheduling:** The decision as to which available process will be executed by the processor

8.4 A process is a program in execution, together with all the state information required for execution.

8.5 The purpose of swapping is to provide for efficient use of main memory for process execution.

8.6 Addresses must be dynamic in the sense that absolute addresses are only resolved during loading or execution.

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8.7 No, if virtual memory is used.

8.8 No.

8.9 No.

8.10 The TLB is a cache that contains those page table entries that have been most recently used. Its purpose is to avoid, most of the time, having to go to disk to retrieve a page table entry.

ANSWERS TO PROBLEMS

8.1 The answers are the same for **(a)** and **(b)**. Assume that although processor operations cannot overlap, I/O operations can.

1 Job:	TAT	=	NT	Processor utilization	= 50%
2 Jobs:	TAT	=	NT	Processor utilization	= 100%
4 Jobs:	TAT	=	(2N – 1)NT	Processor utilization	= 100%

8.2 I/O-bound programs use relatively little processor time and are therefore favored by the algorithm. However, if a processor-bound process is denied processor time for a sufficiently long period of time, the same algorithm will grant the processor to that process because it has not used the processor at all in the recent past. Therefore, a processor-bound process will not be permanently denied access.

8.3 Main memory can hold 5 pages. The size of the array is 10 pages. If the array is stored by rows, then each of the 10 pages will need to be brought into main memory once. If it is stored by columns, then each row is scattered across all ten pages, and each page will have to be brought in 100 times (once for each row calculation).

8.4 The number of partitions equals the number of bytes of main memory divided by the number of bytes in each partition: $2^{24}/2^{16} = 2^8$. Eight bits are needed to identify one of the 2^8 partitions.

8.5 Let *s* and *h* denote the average number of segments and holes, respectively. The probability that a given segment is followed by a hole in memory (and not by another segment) is 0.5, because deletions and creations are equally probable in equilibrium. so with *s* segments in memory, the average number of holes must be *s*/2. It is intuitively reasonable that the number of holes must be less than the number of segments because neighboring segments can be combined into a single hole on deletion.

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8.6 a. Split binary address into virtual page number and offset; use VPN as index into page table; extract page frame number; concatenate offset to get physical memory address

b. (i) 1052 = 1024 + 28 maps to
VPN 1 in PFN 7, ($7 \times 1024+28 = 7196$)
(ii) 2221 = 2×1024 + 173 maps to VPN 2, page fault
(iii) 5499 = 5×1024 + 379 maps to
VPN 5 in PFN 0, ($0 \times 1024+379 = 379$)

8.7 With very small page size, there are two problems: (1) Because very little data is brought in with each page, there will need to be a lot of I/O to bring in the many small pages. (2) The overhead (page table size, length of field for page number) will be disproportionately high.

If pages are very large, main memory will be wasted because the principle of locality suggests that only a small part of the large page will be used.

8.8 9 and 10 page transfers, respectively. This is referred to as "Belady's anomaly," and was reported in "An Anomaly in Space-Time Characteristics of Certain Programs Running in a Paging Machine," by Belady et al, *Communications of the ACM*, June 1969.

8.9 A total of fifteen pages are referenced, the hit ratios are:

N	1	2	3	4	5	6	7	8
Ratio	0/15	0/15	2/15	3/15	5/15	8/15	8/15	8/15

8.10 The principal advantage is a savings in physical memory space. This occurs for two reasons: (1) a user page table can be paged in to memory only when it is needed. (2) The operating system can allocate user page tables dynamically, creating one only when the process is created.

Of course, there is a disadvantage: address translation requires extra work.

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8.11 The machine language version of this program, loaded in main memory starting at address 4000, might appear as:

```
4000       (R1) ← ONE            Establish index register for i
4001       (R1) ← n              Establish n in R2
4002       compare R1, R2        Test i > n
4003       branch greater 4009
4004       (R3) ← B(R1)          Access B[i] using index register R1
4005       (R3) ← (R3) + C(R1)   Add C[i] using index register R1
4006       A(R1) ← (R3)          Store sum in A[i] using index register R1
4007       (R1) ← (R1) + ONE     Increment i
4008       branch 4002
6000-6999  storage for A
7000-7999  storage for B
8000-8999  storage for C
9000       storage for ONE
9001       storage for n
```

The reference string generated by this loop is

$$494944(47484649444)^{1000}$$

consisting of over 11,000 references, but involving only five distinct pages.

8.12 The S/370 segments are fixed in size and not visible to the programmer. Thus, none of the benefits listed for segmentation are realized on the S/370, with the exception of protection. The P bit in each segment table entry provides protection for the entire segment.

8.13 On average, p/2 words are wasted on the last page. Thus the total overhead or waste is w = p/2 + s/p. To find the minimum, set the first derivative to 0.

$$\frac{dw}{dp} = \frac{1}{2} - \frac{s}{p^2} = 0$$

$$p = \sqrt{2s}$$

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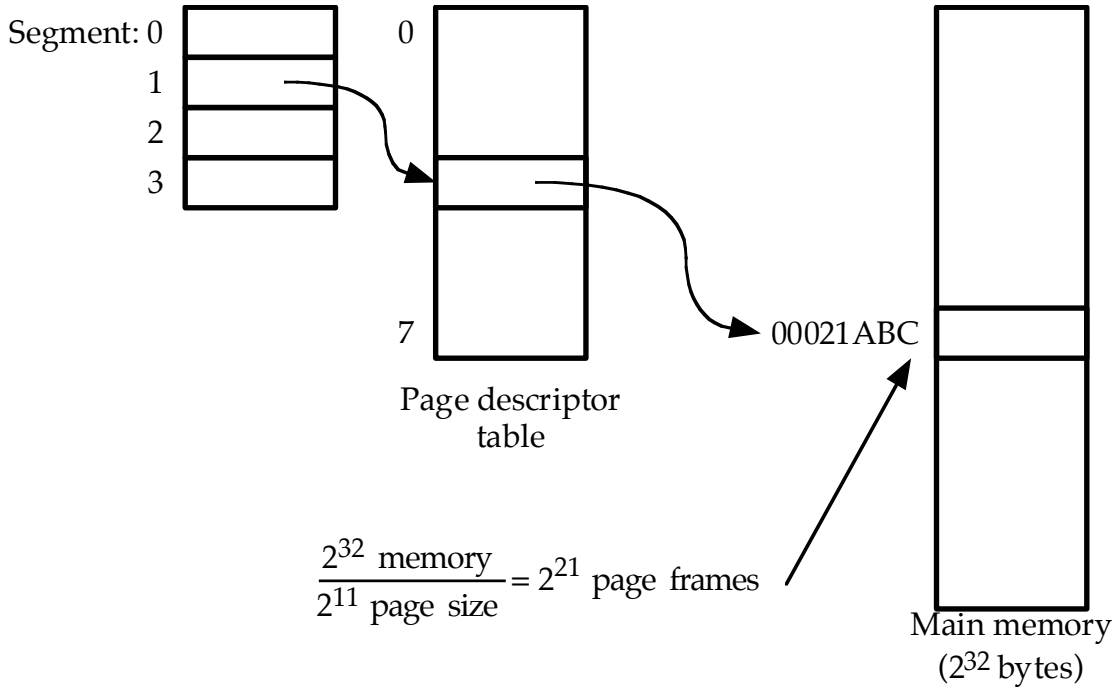
8.14 There are three cases to consider:

Location of referenced word	Probability	Total time for access in ns
In cache	0.9	20
Not in cache, but in main memory	(0.1)(0.6) = 0.06	60 + 20 = 80
Not in cache or main memory	(0.1)(0.4) = 0.04	12ms + 60 + 20 = 12000080

So the average access time would be:

$$\text{Avg} = (0.9)(20) + (0.06)(80) + (0.04)(12000080) = 480026 \text{ ns}$$

8.15 $\frac{2^{32} \text{ memory}}{2^{11} \text{ page size}} = 2^{21} \text{ page frames}$



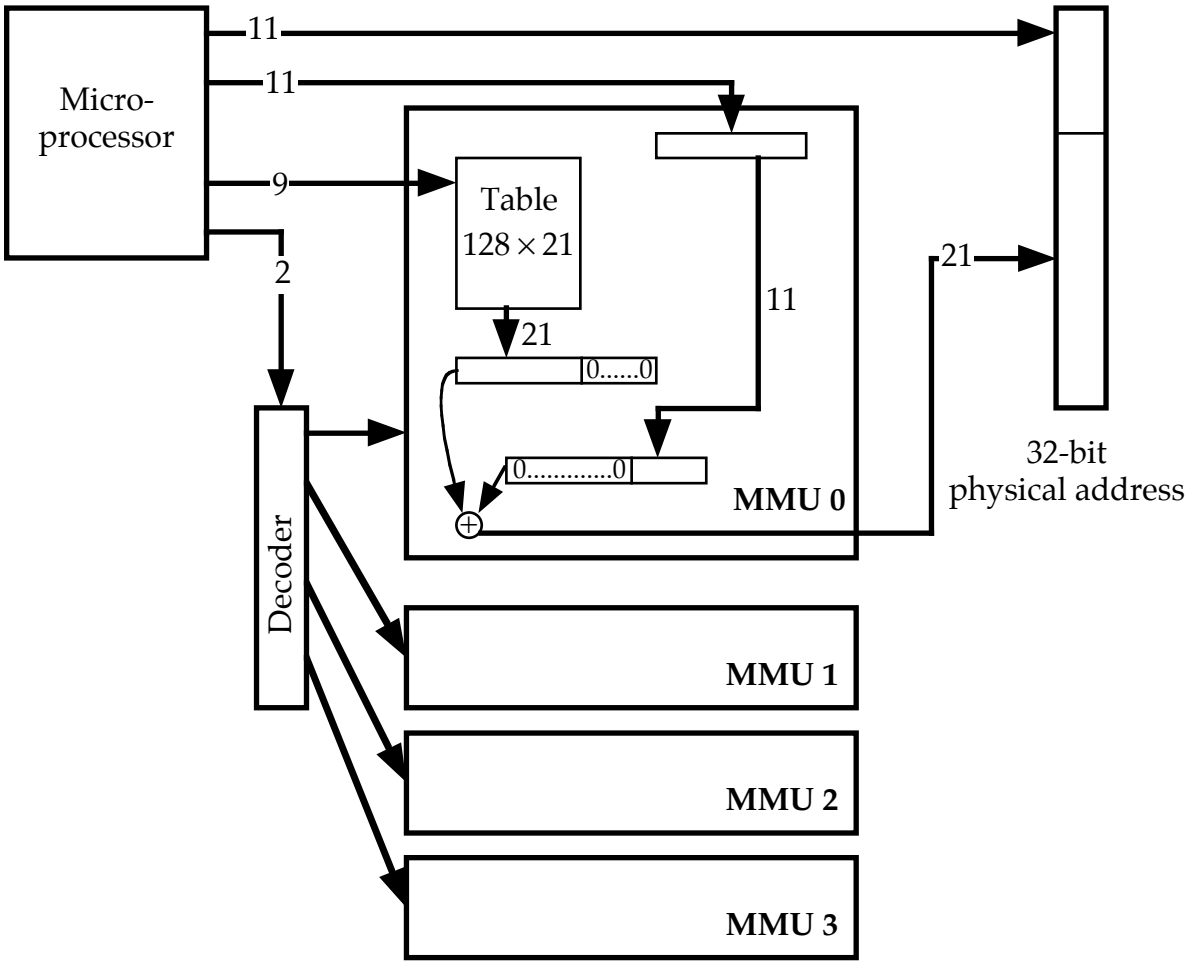
a. $8 \times 2K = 16K$
b. $16K \times 4 = 64K$
c. $2^{32} = 4$ GB

8.16 •The starting physical address of a segment is always evenly divisible by 1048, i.e., its rightmost 11 bits are always 0.
•Maximum logical address space = $2^9 = 512$ segments ($\times 2^{22}$ bytes/segment) = 2^{31} bytes.
•Format of logical address:

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segment number (9)	offset (22)

- Entries in the mapping table: $2^9 = 512$.
- Number of memory management units needed = 4.
- Each 9-bit segment number goes to an MMU; 7 bits are needed for the 128-entry table, the other 2 most significant bits are decoded to select the MMU.
- Each entry in the table is 22 bits.



8.17 a.

page number (5)	offset (11)

b. 32 entries, each entry is 9 bits wide.

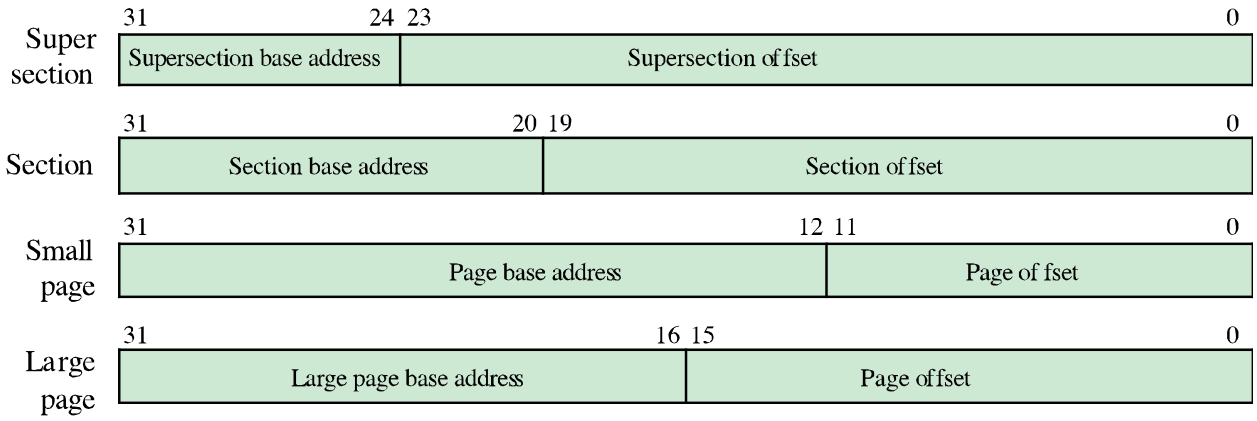
c. If total number of entries stays at 32 and the page size does not change, then each entry becomes 8 bits wide.

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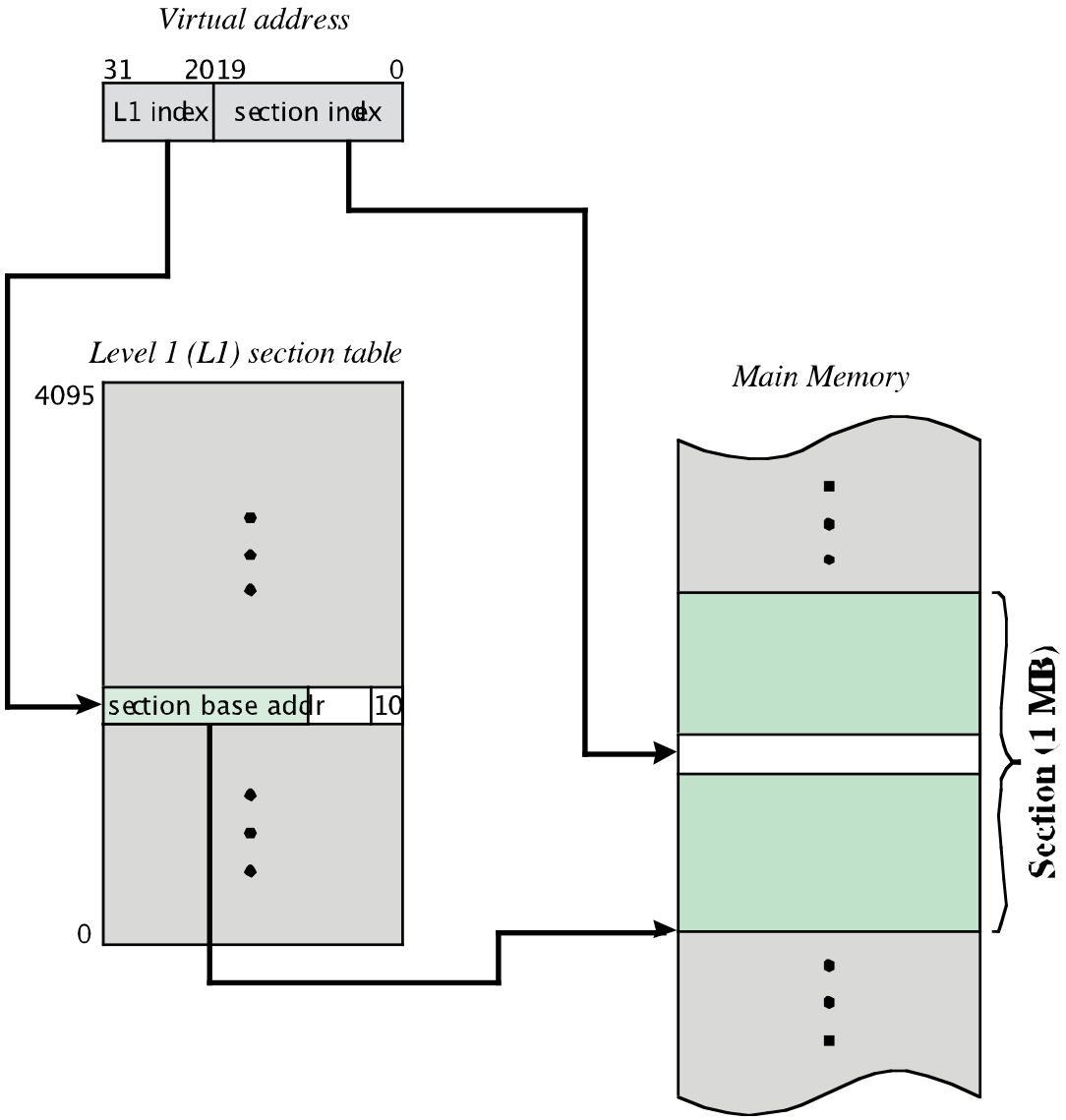
8.18 The system operator can review this quantity to determine the degree of "stress" on the system. By reducing the number of active jobs allowed on the system, this average can be kept high. A typical guideline is that this average should be kept above 2 minutes. This may seem like a lot, but it isn't.

8.19



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8.20



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CHAPTER 9 NUMBER SYSTEMS

ANSWERS TO PROBLEMS

9.1

Base 10	Base 8	Base 6	Base 5	Base 3
1	1	1	1	1
2	2	2	2	2
3	3	3	3	10
4	4	4	4	11
5	5	5	10	12
6	6	10	11	20
7	7	11	12	21
8	10	12	13	22
9	11	13	14	100
10	12	14	20	101
11	13	15	21	102
12	14	20	22	110
13	15	21	23	111
14	16	22	24	112
15	17	23	30	120
16	20	24	31	121
17	21	25	32	122
18	22	30	33	200
19	23	31	34	201
20	24	32	40	202

9.2 $(1.5)_{16} = (1.3125)_{10}$; $(1.4)_{10} = (1.4)_{10}$; $(1.1)_2 = (1.5)_{10}$

9.3 **a.** 134
b. 105
c. 363
d. 185

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9.4 Given a representation of a number x in base n and base n^p, every p digits in the base n representation can be converted to a single base n^p digit. For example, the base 2 representation of decimal 37 is 100101, and the base 8 (2^3) representation is 45. Thus it is easy to convert between base n and base n^p without the intermediate step of converting to base 10.

9.5 **a.** 12 **b.** 3 **c.** 28 **d.** 60 **e.** 42

9.6 **a.** 28.375 **b.** 51.59375 **c.** 682.5

9.7 **a.** 1000000 **b.** 1100100 **c.** 1101111 **d.** 10010001 **e.** 11111111

9.8 **a.** 100010.11 **b.** 11001.01 **c.** 11011.0011

9.9 $1/2^k = 5^k/10^k$

9.10 A BAD ADOBE FACADE FADED

9.11 **a.** 12 **b.** 159 **c.** 3410 **d.** 1662 **e.** 43981

9.12 **a.** 15.25 **b.** 211.875 **c.** 4369.0625 **d.** 2184.5 **e.** 3770.75

9.13 **a.** 10 **b.** 50 **c.** A00 **d.** BB8 **e.** F424

9.14 **a.** CC.2 **b.** FF.E **c.** 277.4 **d.** 2710.01

9.15 **a.** 1110 **b.** 11100 **c.** 101001100100 **d.** 11111.11
e. 1000111001.01

9.16 **a.** 9.F **b.** 35.64 **c.** A7.EC

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CHAPTER 10 COMPUTER ARITHMETIC

ANSWERS TO QUESTIONS

10.1 **Sign–Magnitude Representation**: In an *N*-bit word, the left-most bit is the sign (0 = positive, 1 = negative) and the remaining $N - 1$ bits comprise the magnitude of the number. **Twos Complement Representation**: A positive integer is represented as in sign magnitude. A negative number is represented by taking the Boolean complement of each bit of the corresponding positive number, then adding 1 to the resulting bit pattern viewed as an unsigned integer. **Biased representation**: A fixed value, called the bias, is added to the integer.

10.2 In sign-magnitude and twos complement, the left-most bit is a sign bit. In biased representation, a number is negative if the value of the representation is less than the bias.

10.3 Add additional bit positions to the left and fill in with the value of the original sign bit.

10.4 Take the Boolean complement of each bit of the positive number, then adding 1 to the resulting bit pattern viewed as an unsigned integer.

10.5 When the operation is performed on the n-bit integer -2^{n-1} (one followed by $n - 1$ zeros).

10.6 The **twos complement representation** of a number is the bit pattern used to represent an integer. The **twos complement** of a number is the operation that computes the negation of a number in twos complement representation.

10.7 The algorithm for performing twos complement addition involves simply adding the two numbers in the same way as for ordinary addition for unsigned numbers, with a test for overflow. For multiplication, if we treat the bit patterns as unsigned numbers, their magnitude is different from the twos complement versions and so the magnitude of the result will be different.

10.8 Sign, significand, exponent, base.

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10.9 An advantage of biased representation is that nonnegative floating-point numbers can be treated as integers for comparison purposes.

10.10 **Positive overflow** refers to integer representations and refers to a number that is larger than can be represented in a given number of bits. **Exponent overflow** refers to floating point representations and refers to a positive exponent that exceeds the maximum possible exponent value. **Significand overflow** occurs when the addition of two significands of the same sign result in a carry out of the most significant bit.

10.11 **1.** Check for zeros. **2.** Align the significands. **3.** Add or subtract the significands.
4. Normalize the result.

10.12 To avoid unnecessary loss of the least significant bit.

10.13 **Round to nearest:** The result is rounded to the nearest representable number. **Round toward +∞:** The result is rounded up toward plus infinity. **Round toward −∞:** The result is rounded down toward negative infinity. **Round toward 0:** The result is rounded toward zero.

ANSWERS TO PROBLEMS

10.1

Sign Magnitude:	512	=	0000 0010 0000 0000
	−29	=	1000 0000 0001 1101
Twos Complement:	512	=	0000 0010 0000 0000
	−29	=	1111 1111 1110 0011

10.2 1101011: Because this starts with a leftmost 1, it is a negative number. The magnitude of the negative number is determined by flipping the bits and adding 1:

$$0010100 + 1 = 0010101$$

This is 21, so the original value was −21.

0101101

Because this starts with a leftmost 0, it is a positive number and we just compute the magnitude as an unsigned binary number, which is 45.

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10.3 a. $A = -\left(2^{n-1} - 1\right)a_{n-1} + \sum_{i=0}^{n-2} 2^i a_i$

b. From $-(2^{n-1} - 1)$ through $(2^{n-1} - 1)$

c. **(1)** Add the two numbers as if they were unsigned integers. **(2)** If there is a carry out of the sign position, then add that bit to the first bit position of the result and propagate carries as necessary. This is known as the end-around carry rule. **(3)** An overflow occurs if two positive numbers are added and the result is negative of if two negative numbers are added and the result is positive.

10.4

	sign-magnitude	**ones complement**
Range	$-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$	$-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
Number of representations of 0	2	2
Negation	Complement the sign bit	Complement each bit
Expansion of bit length	Move the sign bit to the new leftmost bit; fill in with zeros	Fill all new bit positions to the left with the sign bit
Subtract B from A	Complement the sign bit of B and add B to A using rules for addition of sign-magnitude numbers	Take the ones complement of B and add it to A

Rules for adding two sign-magnitude numbers:

1. If A and B have the same sign, then add the two magnitudes. If there is a carry out of the last magnitude bit, there is an overflow. If there is no carry the result is the sum of the magnitudes with the same sign bit as A and B.

2. (a) If the magnitude of A equals the magnitude of B, the result is zero; (b) if the magnitude of A is greater than the magnitude of B, then the sign bit of the result is the sign of A, and the magnitude of the result is the magnitude of A minus the magnitude of B. (b) Otherwise, the sign bit of the result is the sign of B, and the magnitude of the result is the magnitude of B minus the magnitude of A.

10.5 The twos complement of the original number.

10.6 a. We can express 2^n as (1 + Z), where Z is an n-bit quantity of all 1 bits. Then, treating all quantities as unsigned integers, we have $(2^n - X) = 1 + Z - X$. But (Z – X) results in the Boolean complement of each bit of *X*. Example:

$$\begin{array}{r} 11111111 \\ -01110100 \\ \hline 10001011 \end{array}$$

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Therefore, (2^n – X) adds one to the quantity formed by taking the Boolean complement of each bit of X, which is how we defined the twos complement of X.

b. In Figure 10.5a, notice that we can subtract *X* or (add –*X*) by moving 16 – *X* positions clockwise. Similarly, in Figure 10.5b, we can subtract *X* or (add –*X*) by moving 2^n – *X* positions clockwise. But the quantity (2^n – *X*) is what we just defined as the twos complement of *X*, which is the twos complement representation of –*X*. So we can subtract *X* by adding –*X*.

10.7 The tens complement is calculated as 10^5 – 13250 = 100000 -13250 = 86750.

10.8 We subtract M – N, where M = 72532 and N = 13250:

M	=	72532
tens complement of N	=	+86750
sum	=	159282
discard carry digit	=	–100000
result	=	59282

10.9

Input	x_{n-1}	0	0	0	0	1	1	1	1
	y_{n-1}	0	0	1	1	0	0	1	1
	c_{n-2}	0	1	0	1	0	1	0	1
Output	z_{n-1}	0	0	1	0	1	0	1	1
	v	0	1	0	0	0	0	1	0

10.10

+6	00000110	–6	11111010	+6	00000110	–6	11111010
+13	00001101	+13	00001101	–13	11110011	–13	11110011
+19	00010011	+7	00000111	–7	11111001	–19	11101101

10.11 Add the twos complement, and check for overflow. For b, we must first sign-extend the second term.

```
  111000          11001100          111100001111          11000011
+ 001101        + 00010010        + 001100001101        + 00011000
  000101          11011110          001000011100          11011000
```

a. **b.** **c.** **d.**

In all cases, the signs of the two numbers to be added are different, so there is no overflow.

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10.12 The overflow rule was stated as follows: If two numbers are added, and they are either both positive or both negative, then overflow occurs if and only if the result has the opposite sign. There are four cases:

- Both numbers positive (sign bit = 0) and no carry into the leftmost bit position: There is no carry out of the leftmost bit position, so the XOR is 0. The result has a sign bit = 0, so there is no overflow.
- Both numbers positive and a carry into the leftmost bit position: There is no carry out of the leftmost position, so the XOR is 1. The result has a sign bit = 1, so there is overflow.
- Both numbers negative and no carry into the leftmost position: There is a carry out of the leftmost position, so the XOR is 1. The result has a sign bit of 0, so there is overflow.
- Both numbers negative and a carry into the leftmost position. There is a carry out of the leftmost position, so the XOR is 0. The result has a sign bit of 1, so there is no overflow.

Therefore, the XOR result always agrees with the presence or absence of overflow.

10.13 An overflow cannot occur because addition and subtraction alternate. As a consequence, the two numbers that are added always have opposite signs, a condition that excludes overflow.

10.14

A	Q	Q_{-1}	M	
0000	1010	0	0101	Initial
0000	0101	0	0101	Shift
1011	0101	0	0101	A ← A − M
1101	1010	1	0101	Shift
0010	1010	1	0101	A ← A + M
0001	0101	0	0101	Shift
1100	0101	0	0101	A ← A − M
1110	0010	1	0101	Shift

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10.15 Using M=010111 (23) and Q = 010011 (19) we should get 437 as the result.

A	Q	Q_{-1}	M	
000000	010111	0	010011	Initial
101101	010111	0	010011	A ← A – M
110110	101011	1	010011	Shift
111011	010101	1	010011	Shift
111101	101010	1	010011	Shift
010000	101010	1	010011	A ← A + M
001000	010101	0	010011	Shift
110101	010101	0	010011	A ← A – M
111010	101010	1	010011	Shift
001101	101010	1	010011	A ← A + M
000110	110101	1	010011	Shift

Answer = 0001 1011 0101 (which is 437)

10.16 An n-digit number in base B has a maximum value of $B^n - 1$. We need to show that the maximum product is less than $B^{2n} - 1$.

$$(B^n - 1)(B^n - 1) = B^{2n} - 2B^n + 1 \leq B^{2n} - 1.$$

The inequality is true if

$$-2B^n + 1 \leq -1 \qquad \text{or} \qquad 1 \leq B^n$$

This is always true for $B \geq 2$ and $n \geq 1$.

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10.17

A	Q	M	
00000000	10010011	1011	Initial
00000001	00100110	1011	Shift
11110110		1011	$A \leftarrow A - M$
00000001	00100110	1011	Restore
00000010	01001100	1011	Shift
11110111		1011	$A \leftarrow A - M$
00000010	01001100	1011	Restore
00000100	10011000	1011	Shift
11111001		1011	$A \leftarrow A - M$
00000100	10011000	1011	Restore
00001001	00110000	1011	Shift
11111100		1011	$A \leftarrow A - M$
00001001	00110000	1011	Restore
00010010	01100000	1011	Shift
00000111		1011	$A \leftarrow A - M$
00000111	01100001	1011	$Q_0 \leftarrow 1$
00001110	11000010	1011	Shift
00000011		1011	$A \leftarrow A - M$
00000011	11000011	1011	$Q_0 \leftarrow 1$
00000111	10000110	1011	Shift
11111100		1011	$A \leftarrow A - M$
00000111	10000110	1011	Restore
00001111	00001100	1011	Shift
00000100		1011	$A \leftarrow A - M$
00000100	00001101	1011	$Q_0 \leftarrow 1$

10.18 The nonrestoring division algorithm is based on the observation that a restoration in iteration I of the form $A(I) \leftarrow A(I) + M$ is followed in iteration (I + 1) by the subtraction $A(I+1) \leftarrow 2A(I) - M$. These two operations can be combined into a single operation: $A(I+1) \leftarrow 2A(I) + M$.

10.19 False. For a negative quotient, truncation yields a larger number.

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10.20 Divisor = 13 = $(001101)_2$ is placed in M register.
Dividend = –145 = $(111101101111)_2$ is placed in A and Q registers

A	Q	M	
111101	101111	001101	Initial
111011	011110		Shift
001101			Add
001000			
111011	011110		Restore
110110	111100		Shift
001101			Add
000011			
110110	111100		Restore
101101	111000		Shift
001101			Add
111010	111001		$Q_0 \leftarrow 1$
110101	110010		Shift
001101			Add
000110			
110101	110010		Restore
101011	100100		Shift
001101			Add
111000	100101		$Q_0 \leftarrow 1$
110001	001010		Shift
001101			Add
111110	001011		$Q_0 \leftarrow 1$

Remainder = $(111110)_2$ = –2
Quotient = twos complement of 001011 = $(110101)_2$ = –11

10.21 **a.** Planck's constant:

$$6.63 \times 10^{-27} \rightarrow 0.\underbrace{00000000000000000000000000663}_{29}$$

Avogadro's number:

$$6.02 \times 10^{23} \rightarrow \underbrace{602000000000000000000000}_{24}.0$$

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To represent the approximation of Planck's constant 29 radix-10 fractional digits are needed, while representing the approximation of Avogadro's number requires 24 integer decimal digits. To represent the approximations of both Planck's constant and Avogadro's number in a fixed-point number format, 29 + 54 = 53 radix-10 digits are needed.

b. In the considered radix-10 biased representation for the exponent (such that $E_{biased} = E + 50$), the exponent of both Planck's constant and Avogadro's number can be represented using 2 digits, because 27+50 = 23 and 23+50 = 73. To represent the significands, 3 radix-10 digits are needed. Therefore, to represent the approximations of both Planck's constant and Avogadro's number in a floating-point radix- 10 number format, 3 + 2 = 5 decimal digits are needed.

10.22 **a.** $b^{X-q}(1 - b^{-p})$, b^{-q-p}

b. $b^{X-q}(1 - b^{-p})$, b^{-q-1}

10.23 **a.** 1 10000001 01000000000000000000000
b. 1 10000001 10000000000000000000000
c. 1 01111111 10000000000000000000000
d. $384 = 110000000 = 1.1 \times 2^{1000}$
Change binary exponent to biased exponent:
127 + 8 = 135 = 10000111
Format: 0 10000111 00000000000000000000000
e. $1/16 = 0.0001 = 1.0 \times 2^{-100}$
127 – 4 = 123 = 01111011
Format: 0 01111011 00000000000000000000000
f. $-1/32 = -0.00001 = -1.0 \times 2^{-101}$
127 – 5 = 122 = 01111010
Format: 0 01111010 00000000000000000000000

10.24 **a.** –28 (don't forget the hidden bit)
b. 13/16 = 0.8125
c. 2

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10.25 In this case, the exponent has a bias of 3. Special cases are shaded in the table. The first shaded column contains the denormalized numbers. It is worthwhile to study this table to get a feel for the distribution and spacing of numbers represented in this format.

sign bit and significand	Exponent							
	000	001	010	011	100	101	110	111
0 000	0	0.25	0.5	1	2	4	8	+∞
0 001	0.03125	0.28125	0.5625	1.125	2.25	4.5	9	NaN
0 010	0.0625	0.3125	0.625	1.25	2.5	5	10	NaN
0 011	0.09375	0.34375	0.6875	1.375	2.75	5.5	11	NaN
0 100	0.125	0.375	0.75	1.5	3	6	12	NaN
0 101	0.15625	0.40625	0.8125	1.625	3.25	6.5	13	NaN
0 110	0.1875	0.4375	0.875	1.75	3.5	7	14	NaN
0 111	0.21875	0.46875	0.9375	1.875	3.75	7.5	15	NaN
1 000	−0	−0.25	−0.5	−1	−2	−4	−8	−∞
1 001	−0.03125	−0.28125	−0.5625	−1.125	−2.25	−2.5	−9	NaN
1 010	−0.0625	−0.3125	−0.625	−1.25	−2.5	−5	−10	NaN
1 011	−0.09375	−0.34375	−0.6875	−1.375	−2.75	−5.5	−11	NaN
1 100	−0.125	−0.375	−0.75	−1.5	−3	−6	−12	NaN
1 101	−0.15625	−0.40625	−0.8125	−1.625	−3.25	−6.5	−13	NaN
1 110	−0.1875	−0.4375	−0.875	−1.75	−3.5	−7	−14	NaN
1 111	−0.21875	−0.46875	−0.9375	−1.875	−3.75	−7.5	−15	NaN

10.26

a. $1.0 = +1/16\times16^{1} =$ 0 100 0001 0001 0000 0000 0000 0000 0000

b. $0.5 = +8/16\times16^{0} =$ 0 100 0000 1000 0000 0000 0000 0000 0000

c. $1/64 = +4/16\times16^{-1} =$ 0 011 1111 0100 0000 0000 0000 0000 0000

d. $0.0 = +0\times16^{-64} =$ 0 000 0000 0000 0000 0000 0000 0000 0000

e. $-15.0 = -15/16\times16^{1} =$ 1 100 0001 1111 0000 0000 0000 0000 0000

f. $5.4\times10^{-79} \approx +1/16\times16^{-64} =$ 0 000 0000 0000 0000 0000 0000 0000 0000

g. $7.2\times10^{75} \approx 1\times16^{63} =$ 0 111 1111 1111 1111 1111 1111 1111 1111

h. $65535 = 16^{4}-1 =$ 0 100 0100 1111 1111 1111 1111 0000 0000

10.27 Step 1: Sign positive

Step 2: Extract the exponent $(5B)_{16}$ and subtract the bias $(40)_{16}$, yielding

$$(1B)_{16} = 27$$

Step 3: The significand $(CA\ 0000)_{16} = 12/16 + 10/256 = 0.7890625$. The decimal result is 0.7890625×16^{27}.

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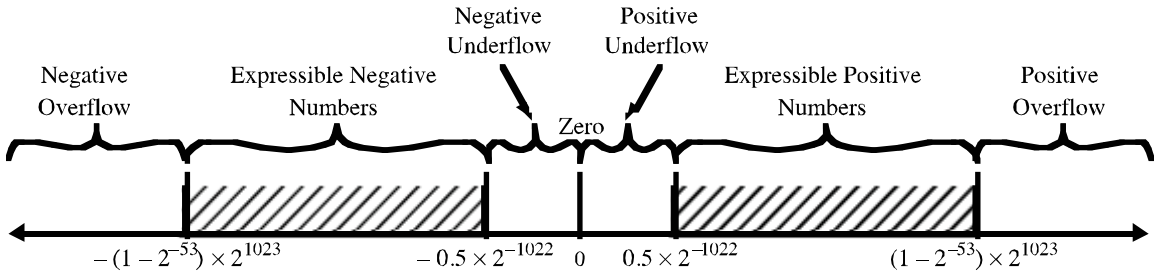
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10.28 The base is irrelevant

a. Bias = $2^{6-1} = 2^5 = 32$

b. Bias = $2^{7-1} = 2^6 = 64$

10.29



10.30 **a.** 1. Express the number in binary form: 1011010000 (normalize to 1.1bbbb)

2. Normalize the number into the form 0.1bbbbbbbbbbbbb
   0.1011010000×2^k where k = 10(base10) or 1010(base2)
   $0.1011010000 \times 2^{(1010)}$
   Once in normalized form every number will have a 1 after the decimal point. We do not need to store this number; it is implicit. Therefore in the Significand field we will store 01101000000000000000.
3. For the 8-bit exponent field, a bias of 128 is used. Add the bias to the exponent and store the answer: 1010 + 10000000 = 1001010
4. Sign bit = 1
5. Result = 1 1001010 01101000000000000000

b. We have 0.645 = 0.101001...; therefore the significand is 01001 (the first 1 is implicit). The sign = 0, and the exponent = 0. Result: 0 0000000 01001000000000000000

10.31 There are 2^{32} different bit patterns available. However, because of special cases, not all of these bit patterns represent unique numbers. In particular, an exponent of all ones together with a nonzero fraction is given the value NaN, which means *Not a Number*, and is used to signal various exception conditions. Because the fraction field is 23 bits, the number of nonzero fractions is $2^{23} - 1$. The sign bit may be 0 or 1 for this case, so the total number of NaN values is $2^{24} - 2$. Therefore, the number of different numbers that can be represented is $2^{32} - 2^{24} + 2$. This number includes both plus and minus zero and plus and minus infinity. If we exclude minus zero and plus and minus infinity, then the total is $2^{32} - 2^{24} - 1$.

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10.32 We have 0.4×2^0. Because 0.4 is less than 0.5, this is not normalized. Thus, we rewrite as

$$0.4 = 0.8 \times 2^{-1}$$

Next, convert 0.8 to binary, we have repeating binary number: 0.110011001100... The closest we can get (7 bits) is 0.1100110. Converting this back to decimal, we have

$$(1/2 + 1/4 + 1/32 + 1/64) \times 2^{-1} = 0.3984375$$

The relative error is $\dfrac{0.4 - 0.3984375}{0.4} = 0.0039$

10.33 $E_A = \dfrac{A - A'}{A}$

Truncation: $E_A = \dfrac{1.427 - 1.42}{1.427} = 0.0049$

Rounding: $E_A = \dfrac{1.427 - 1.43}{1.427} = -0.0021$

10.34 Cancellation reveals previous errors in the computation of X and Y. For example, if ε is small, we often get poor accuracy when computing $f(x + \varepsilon) - f(x)$, because the rounded calculation of $f(x + \varepsilon)$ destroys much of the information about ε. It is desirable to rewrite such formulas as $\varepsilon \times g(x,\varepsilon)$, where $g(x,\varepsilon) = \dfrac{f(x+\varepsilon) - f(x)}{\varepsilon}$ is first computed symbolically. Thus, if $f(x) = x^2$, then $g(x,\varepsilon) = 2x + \varepsilon$; if $f(x) = \sqrt{x}$, then $g(x,\varepsilon) = \dfrac{1}{\sqrt{x + \varepsilon} + \sqrt{x}}$.

10.35 We have

$$E_A = \frac{A - A'}{A} = 1 - \frac{A'}{A}$$

$A' = A(1 - E_A)$
$B' = B(1 - E_B)$
$A'B' = AB(1 - E_A)(1 - E_B) = AB[1 - (E_A + E_B) + E_A E_B]$
$\approx AB[1 - (E_A + E_B)]$

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The product term $E_A E_B$ should be negligible in comparison to the sum. Consequently

$$E_{AB} = E_A + E_B.$$

10.36 **a.** $E_A = \dfrac{0.22288 - 0.2228}{0.2228} = 0.00036$

$E_B = \dfrac{0.22211 - 0.2221}{0.22211} = 0.00045$

b. $C = A - B = 0.00077$

$C' = A' - B' = 0.0007$

$E_C = \dfrac{0.00077 - 0.0007}{0.00077} = 0.09$

10.37 **a.** $(2.50000 \times 10^{-60}) \times (3.50000 \times 10^{-43}) = 8.75000 \times 10^{-103} \rightarrow 0.00088 \times 10^{-99}$

The otherwise exact product underflows and must be denormalized by four digits. The number then requires rounding.

b. $(2.50000 \times 10^{-60}) \times (3.50000 \times 10^{-60}) = 8.75000 \times 10^{-120} \rightarrow 0.0$

The intermediate result falls below the underflow threshold and must be set to zero.

c. $(5.67834 \times 10^{-97}) - (5.67812 \times 10^{-97}) = 2.20000 \times 10^{-101} \rightarrow 0.02200 \times 10^{-99}$

This example illustrates how underflowed sums and differences of numbers in the same format are always free from rounding errors.

10.38 **a.** The exponents are equal. Therefore the mantissas are added, keeping the common exponent, and the sum is renormalized if necessary.

$5.566 \times 10^3 + 7.777 \times 10^3 = 1.3343 \times 10^3 \approx 1.334 \times 10^3$

b. The exponents must be equalized first.

$3.344 \times 10^1 + 8.877 \times 10^{-2} = 3.344 \times 10^1 + 0.008877 \times 10^1 = 3.352877 \times 10^1 \approx 3.352 \times 10^1$

10.39 **a.** $7.744 \times 10^{-3} - 6.666 \times 10^{-3} = 1.078 \times 10^{-3}$

b. $8.844 \times 10^{-3} - 2.233 \times 10^{-1} = 0.08844 \times 10^{-1} - 2.233 \times 10^{-1} = -2.14456 \times 10^{-1} \approx -2.144 \times 10^{-1}$

10.40 **a.** $2.255 \times 10^1 \times 1.234 \times 10^0 = 2.58267 \times 10^1 \approx 2.582 \times 10^1$

b. $8.833 \times 10^2 \div 5.555 \times 10^4 = 1.590 \times 10^{-2}$

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